RESEARCH ARTICLE

MapSafe Mobile: Embedding Geospatial Data Sovereignty into Mobile Field Data Collection and Secure Community Sharing

Pankajeshwara Sharma[a]

[a]Universal College of Learning, Palmerston North, New Zealand

ARTICLE HISTORY
Compiled October 9, 2026

ABSTRACT
Mobile GIS makes it possible to record precise environmental observations at the point of collection, but decisions about spatial disclosure, authorised recipients, and integrity evidence are often postponed until after synchronisation. Sensitive coordinates may therefore cross an external trust boundary before the data custodian can protect them. This paper asks how geoprivacy and controlled sharing can be incorporated into an established mobile field-data workflow. We present MapSafe Mobile, an extension of the open-source NextGIS Mobile application that preserves the authoritative observation, creates halo-masked or H3-aggregated alternatives, encrypts a selected dataset once for chosen OpenPGP recipients, exchanges public keys and protected resources through a NextGIS Web community, and optionally records a filename-bound digest in an externally signed EVM transaction. The approach is demonstrated with the North Whangārei biodiversity case and evaluated through a live four-account community acceptance test, physical-device measurements over ten performance datasets, and first-author functional validation of 24 prepared task trials. On a Samsung Galaxy A14, the 1,000-point rich dataset required median times of 5.96 s for complete masking, 4.24 s for signed encryption, and 3.52 s for verified decryption; corresponding 2,000-point stress-case times were 11.52, 6.14, and 4.69 s. The results show that capture-time protection is computationally practical on a modest handset and that spatial representation, recipient access, community permissions, and integrity evidence can be kept as separate decisions. MapSafe Mobile extends earlier browser and desktop MapSafe workflows by providing a practical, open-source foundation for capture-time geoprivacy and community-governed sharing in mobile GIS.

KEYWORDS
Indigenous Data Sovereignty; mobile GIS; geoprivacy; geomasking; H3; OpenPGP; secure spatial data sharing

1. Introduction

Mobile devices support environmental monitoring, citizen science, participatory sensing, and community mapping, but they also collect precise coordinates, photographs, timestamps, and notes at increasing scale. Users may not fully understand what is collected, how it is combined, or how it will be used (McKenzie, Keßler, and Andris, 2019). Fine-grained location records can reveal lifestyles, political and religious affiliations, identities, sensitive species, culturally important places, and community

CONTACT Pankajeshwara Sharma. Email: pankajeshwara@gmail.com; ORCID: https://orcid.org/0000-0001-9159-8332

relationships (Christin, 2016; Christin et al., 2011; Pyrgelis, De Cristofaro, and Ross, 2016). Precision increases disclosure risk, while aggregation and geomasking involve unavoidable privacy–utility trade-offs (Keßler and McKenzie, 2018; Kim, Edemacu, and Jang, 2022). These risks also extend beyond individuals because contextual inference, repeated releases, and apparently harmless datasets can expose communities (Masinde et al., 2023).

Although previous mobile and distributed systems have aggregated or transformed locations locally before transmission, most address individual privacy or location-based service queries rather than the complete management of a field dataset (Chow, Mokbel, and Liu, 2006; Meftah, Rouvoy, and Chrisment, 2021; Pyrgelis, De Cristofaro, and Ross, 2016; Ye et al., 2019). Protection is therefore often applied only after synchronisation or publication, when precise observations may already have crossed an external trust boundary and may be difficult to recover. The problem is not only how to anonymise data before release, but how to preserve community authority over representation, recipients, and subsequent sharing from the moment observations are created. Although location privacy has received substantial research attention, practical mobile crowdsourcing systems still provide limited protection during data collection and publication (Boukoros et al., 2019; Kim, Edemacu, and Jang, 2022).

The timing of protection is important. In many workflows, precise observations are collected on a phone and protected only after they have been synchronised to a desktop computer, institutional server, or cloud platform. By then, the original coordinates have already left the device and may exist in systems that the collector does not fully control. The field device is therefore an important point at which the collector can still decide what geographic detail should be disclosed, who may recover the original data, and what evidence should accompany a shared dataset.

These decisions are particularly important in Indigenous and community-based environmental monitoring. Mobile mapping has helped communities document fishing grounds, natural resources, ecological conditions, and culturally important places (Paul et al., 2016). Applications such as *CyberTracker* and *GeoKeeper* also show how mobile tools can support Indigenous knowledge, guardianship, and offline environmental monitoring (Ens, 2012; Kwusen Research & Media, 2026). However, a mobile data-collection application does not, by itself, provide geoprivacy or data sovereignty. Exact coordinates may still be exposed when data are synchronised, shared, or reused outside their intended context.

Indigenous data sovereignty provides a governance basis for addressing this problem. It recognises the rights of Indigenous peoples to govern the collection, ownership, access, interpretation, and use of data about their peoples, lands, cultures, and resources (Kukutai and Taylor, 2016). The CARE Principles similarly emphasise collective benefit, authority to control, responsibility, and ethics throughout the data lifecycle (Carroll et al., 2020). These responsibilities begin when data are created, not only when they are later stored or published. Recent research therefore calls for data-collection practices that give Indigenous communities greater control over what is recorded, how it is managed, and who may use it (Engstrom et al., 2024; Marriott et al., 2025). Technology cannot confer sovereignty or legitimate authority, but it can help authorised custodians put their governance decisions into practice.

Geoprivacy research offers several relevant safeguards, including geomasking, spatial aggregation, encryption, controlled sharing, and integrity verification. Earlier versions of *MapSafe* brought these functions into browser and desktop GIS workflows so that a data owner could prepare different spatial representations and protect sensitive files before sharing them (Sharma, Martin, and Swanlund, 2023; Sharma et al.,

2024; Sharma, Young, and Dubrava, 2025). These approaches are useful after a dataset reaches a browser or desktop GIS, but they do not close the gap between collecting a sensitive observation in the field and protecting it. Mobile GIS applications support data capture, mapping, and synchronisation, yet they rarely combine local anonymisation, recipient-controlled encryption, community exchange, and independent integrity checking in one field workflow.

This study therefore asks: *How can spatial anonymisation, recipient-controlled encryption, community exchange, and integrity verification be incorporated into a mobile GIS workflow before sensitive observations leave the collecting device?* We describe this aim as *capture-time sovereignty*: enabling an authorised data custodian to make representation, access, and integrity decisions at or soon after data collection, rather than trying to recover control after precise data have already been synchronised elsewhere.

To address this question, we developed *MapSafe Mobile* as an extension of the open-source *NextGIS Mobile* platform. The application preserves the authoritative source while allowing the custodian to create either halo-masked points or an H3 hexagonal aggregate. A selected dataset can be encrypted once for one or more OpenPGP recipients, avoiding a shared dataset passphrase and allowing each authorised recipient to use their own protected private key. *NextGIS Web* supports community membership, public-key discovery, anonymised-layer exchange, encrypted-package distribution, and resource access-control lists. An optional EVM workflow records a filename-bound digest for later comparison. On access, the application can download or open a protected package, verify its local and recorded digest, decrypt it for an authorised recipient, and display the recovered dataset. Spatial transformation and cryptographic processing occur locally; community exchange and blockchain operations require connectivity.

The paper makes four contributions. First, it defines capture-time sovereignty as a practical design goal for mobile geoprivacy. Second, it implements an end-to-end safeguard and access workflow within a mobile GIS. Third, it separates two decisions that are often conflated: which spatial representation may be shared and which recipients may access an encrypted file. Fourth, it evaluates the approach through a North Whangārei biodiversity-monitoring case study, a live four-account community acceptance test, physical-device performance measurements, and a usability-informed effectiveness assessment.

The remainder of the paper is organised as follows. Section 2 reviews Indigenous geospatial data sovereignty in community-based and mobile environmental monitoring and introduces the case context. Section 3 presents the case actors, disclosure decisions, system design, implementation, workflows, and technical evaluation. Section 4 describes the usability-informed design and effectiveness assessment. Section 5 discusses the implications, limitations, and relationship to earlier MapSafe implementations, and Section 6 concludes the paper.

2. Indigenous geospatial data sovereignty in mobile fieldwork

Indigenous Data Sovereignty concerns the right of Indigenous peoples to govern data about their peoples, territories, resources, cultures, and ways of life, whether those data are collected by the community or by others (Kukutai and Taylor, 2016; Rainie et al., 2019; Walter et al., 2021). In Aotearoa New Zealand, Māori data-sovereignty scholarship places this authority within relationships among people, knowledge, and place rather than treating data as a resource controlled by whoever possesses a database (Kukutai and Cormack, 2019; Sporle, Hudson, and West, 2020; Te Mana Raraunga,

2018). The CARE Principles similarly extend governance across the data lifecycle through collective benefit, authority to control, responsibility, and ethics (Carroll et al., 2020, 2021). These principles are especially important where open-data, big-data, and cloud practices can separate information from its origins and intended purpose (Calac and Mackey, 2025; Cormack and Kukutai, 2022; Lovett et al., 2019).

Biodiversity monitoring makes this relationship between data and place particularly visible. Indigenous peoples and local communities hold long-standing knowledge of ecological change and increasingly contribute observations used in conservation and environmental decision-making (Forster, 2023; Reyes-García et al., 2022). In Aotearoa, responses to kauri dieback and other threats have shown the importance of Māori leadership, cross-cultural collaboration, and monitoring methods that connect surveillance with local management authority (Greenaway et al., 2023; Hill et al., 2021; Lambert et al., 2018; Waipara et al., 2013). Community-based forest-health monitoring can strengthen local learning and action, but participation in collection does not automatically determine who governs the resulting records or how they may be reused (Lyver et al., 2017; Massarella, Sallu, and Ensor, 2020; Wilson, 2025).

Mobile technology has made such monitoring more practical. Smartphones combine positioning, photographs, timestamps, notes, local storage, and network communication in one field device. Participatory smartphone mapping has supported documentation of fishing grounds and resources, while CyberTracker has enabled Indigenous land and sea managers to record environmental work using locally usable interfaces (Ens, 2012; Paul et al., 2016). Reviews nevertheless find that successful smartphone monitoring depends more on people, partnerships, continuing costs, and connections to action than on the data-collection technology alone (Andrachuk et al., 2019; Cochrane and Corbett, 2020; Wiseman and Bardsley, 2016). A mobile application can therefore support community authority, but cannot create legitimate authority or self-determination by itself (Reyes-García et al., 2022; Wilson, 2025).

The geospatial character of these records creates an additional risk. Maps can support land claims, environmental management, and counter-mapping, but they can also expose culturally significant places or enable knowledge to be removed from its proper context (Duckham and Ho, 2020; Kidd, 2019; Wainwright and Bryan, 2009). Indigenous GIS scholarship has consequently called for infrastructures that let communities determine what is represented, who may receive it, and where it may circulate (Briggs et al., 2020; Olson, Hackett, and DeRoy, 2016; Reid and Sieber, 2022). These concerns extend to mobile fieldwork: an exact coordinate may be saved together with revealing attributes and later synchronised to a platform outside the collector's direct control. Research on participatory sensing shows that location disclosure can persist even where applications offer general privacy controls (Boukoros et al., 2019; Christin, 2016; Christin et al., 2011; Kim, Edemacu, and Jang, 2022).

The earlier cloud-based MapSafe study translated this governance problem into several technical requirements for Māori biodiversity data: provide an obfuscated representation for an approved limited purpose, protect the original before cloud storage, allow several sovereign or collaborating parties to use their own keys, and retain independently checkable integrity evidence (Sharma et al., 2024). This is not a claim that anonymisation makes a dataset harmless. Geomasking can preserve useful spatial patterns, but contextual information or repeated releases may still enable inference or re-identification (Ajayakumar, Curtis, and Curtis, 2019; Cassa, Wieland, and Mandl, 2008; Hojati et al., 2021; Keßler and McKenzie, 2018; Zimmerman and Pavlik, 2008). Nor do platform permissions alone ensure sovereignty when a service provider can inspect plaintext or control the encryption keys. Representation, cryptographic ac-

cess, platform authorisation, and governance therefore remain related but distinct safeguards.

The browser, cloud, and desktop versions of MapSafe progressively brought anonymisation, encryption, controlled sharing, and integrity checking into common GIS environments (Sharma, Martin, and Swanlund, 2023; Sharma et al., 2024; Sharma, Young, and Dubrava, 2025). They protect a dataset once it is opened in a browser, uploaded through a protected web workflow, or handled in desktop GIS. Mobile field collection exposes an earlier boundary: the precise observation already exists on the handset and may be synchronised before the custodian reaches those later tools. MapSafe Mobile addresses this timing gap by placing the protection decision inside the field-data application, allowing local processing before external transfer. Its contribution is therefore an improvement in when and where established controls can be applied, rather than a claim that a phone replaces community governance.

This mobile extension also changes the practical meaning of control. The data guardian can retain the authoritative observation, inspect an anonymised derivative, select recipients, and protect a release while still working with the field dataset. Offline operation can delay reliance on an external service, while later community exchange can remain subject to purpose-specific permissions. This approach is consistent with research that treats Indigenous data governance as an active stewardship process spanning collection, storage, use, sharing, and reuse, rather than as a consent decision made only at publication (Bowen and Hinze, 2022; Carroll et al., 2020, 2021; Lovett et al., 2019; Reyes-García et al., 2022).

To make this capture-time problem concrete, the paper reuses the North Whangārei infected-tree dataset and Biodiversity Management Area context documented in the earlier cloud study (Sharma et al., 2024). The dataset and case context are not fictional; their reuse provides continuity for examining how the same confidentiality and governance requirements change when observations originate in a mobile GIS. The present study evaluates a mobile implementation rather than reporting new ecological observations, and its technical controls do not replace the governance practices of the relevant BMAs. Section 3.1 introduces the case actors and tested disclosure decisions, while the remainder of Section 3 explains their implementation.

3. MapSafe Mobile architecture and workflow

MapSafe Mobile is integrated into the existing *NextGIS* Mobile host rather than replacing its normal field-data workflow. The application starts with the standard *NextGIS* introduction, account, and map interface; Steven opens MapSafe from the host application's menu when safeguarding or access functions are required. This preserves the familiar process for collecting observations and makes MapSafe an explicit governance step between field capture and external sharing.

The architecture in Fig. 1 separates three boundaries. Spatial transformation, encryption, hashing, and decryption are performed on the *Android* device. *NextGIS* Web supplies authenticated accounts, group-scoped public-key exchange, access-controlled vector resources, and storage of encrypted packages. An EVM-compatible blockchain can independently bind an encrypted package filename to its SHA-256 digest. No plaintext dataset, private key, passphrase, or *OpenPGP* session key is written on-chain; nevertheless, because the encrypted filename is public, owners should use neutral package names that do not disclose sensitive places or subjects. The cloud service can therefore distribute protected material without receiving the *OpenPGP* private keys required

to recover it.

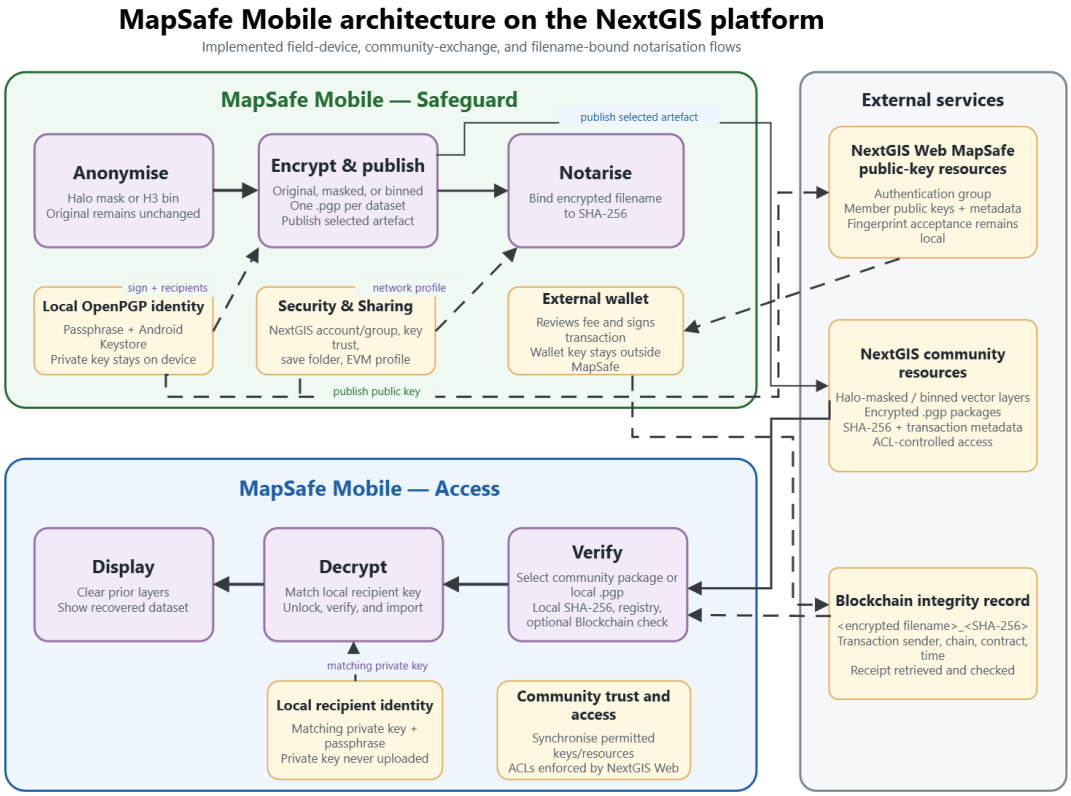
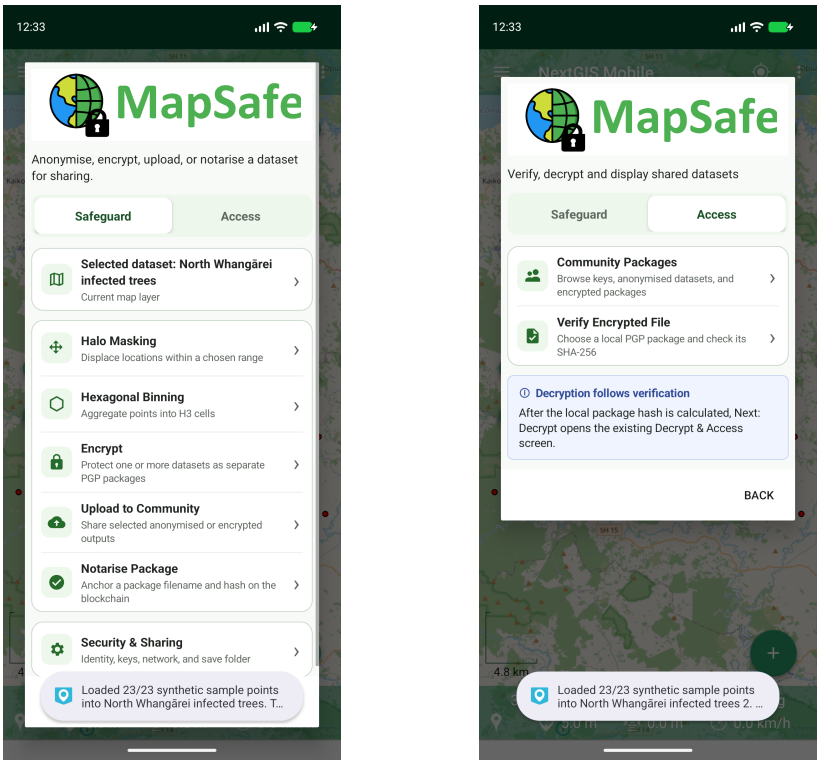


Figure 1.: MapSafe Mobile architecture. Sensitive processing occurs on the field device; NextGIS Web supports community membership, public-key exchange, and protected-resource distribution; an optional EVM record independently binds the encrypted package filename to its digest.

The mobile interface presents *Safeguard* and *Access* as two tabs in one MapSafe panel (Fig. 2). Safeguard provides direct entry to halo masking, hexagonal binning, encryption, community upload, notarisation, and security settings. Access begins with community packages or a locally selected encrypted file and then supports verification, decryption, and display of the recovered dataset. Opening an option launches its focused activity, while the tabs return to the same vertical position and preserve a compact, consistent workflow overview. The stages shown within those activities are orientation cues rather than a compulsory wizard: *Stop* lets Steven or an authorised recipient end after a useful output, whereas *Next* continues to the following safeguard or access stage. A compatible point-vector layer is required before a spatial anonymisation operation, while Access does not require a preselected map layer because its first step obtains an encrypted package.

For practical use and reproducibility, a step-by-step help guide is available at `https://sharmapn.github.io/MapSafeMobile/`. The guide explains the Safeguard, Access, community-sharing, verification, and decryption workflows. It also provides the MapSafe Mobile APK for controlled research and field testing outside Google Play at `https://sharmapn.github.io/mapsafemobile/downloads/MapSafeMobile-3.2.1.apk`. The open-source MapSafe Mobile code is available at `https://github.com/sharmapn/nextgis_mobile_android`.



(a) Safeguard tab and its direct entry points.

(b) Access tab for community or local packages.

Figure 2.: The compact MapSafe workflow panel. A two-state tab control separates safeguard operations from access operations while keeping both views aligned within the same interface.

3.1. Users, disclosure decisions, and community preparation

The case introduced in Section 2 distinguishes Steven's custodial responsibilities, the BMAs' sovereign authority, Amber's purpose-specific research access, and the exclusion of a person with no authorised purpose. The representations and access assigned to each of the four identities in the tested North Whangārei release are recorded in Table 1.

Steven holds the dataset and initiates safeguarding as its custodian, while the *BMAs* retain sovereign authority over its use. In the tested release, that authority is operationalised by addressing the protected original to the BMA representative and limiting Amber to the anonymised alternatives required for her research purpose. Representation and access remain independent, so community membership or organisational standing alone does not confer access to exact coordinates.

NextGIS administration is intentionally separated from field operation. Before fieldwork, an authorised administrator creates or selects the Web GIS, configures *NextGIS* ID/team membership, creates the relevant authentication group, and applies resource *access-control lists* (ACLs) (NextGIS, 2026a,b). Figure 3 shows the four accounts prepared for the Community A evaluation: *Steven–Data Custodian, Amber–Researcher, BMA Representative,* and *External Test User*. These descriptive profile names identify their roles in the case study; they do not themselves grant access to community resources. Team membership establishes the collaboration context, while the Web

Table 1.: Access assigned to each identity in the tested North Whangārei community release. These release-specific permissions do not constitute a permanent hierarchy of authority.

Participant and role	**Representation**	**Purpose**	**Access protection**
Steven – Data Custodian	Original and derived representations	Field stewardship, safeguarding, and controlled release	Local custody, owner ACLs, and a self-recipient OpenPGP envelope
BMA Representative	Encrypted original and authorised anonymised layers	Sovereign oversight and authorised management of precise observations	Package ACLs and an OpenPGP recipient envelope addressed to the BMA representative
Amber – Researcher	Halo-masked points and hexagonal aggregate	Purpose-specific analysis of approximate patterns and regional distribution	Community vector-resource ACLs; absent from the encrypted-original recipient set
Alex (External User)	None	Negative-control identity with no authorised purpose	Excluded by resource ACLs and absent from the OpenPGP recipient set

GIS group and resource ACLs determine which anonymised layers or encrypted packages each account can discover and download. Access to the plaintext inside an encrypted package additionally requires the user's public key to have been selected during *OpenPGP* encryption. Invitation acceptance, account recovery, compromised-account response, membership removal, and storage administration are best completed through the *NextGIS* Web browser interface. In the field, *MapSafe Mobile* consumes an existing authenticated account and group; group creation remains available only when the signed-in account has the necessary server permission.

Thus, the first community connection is normally prepared in *NextGIS* Web rather than created entirely inside MapSafe. The mobile application reuses the authenticated NextGIS account, lets the user select an existing group, publishes only the user's public key, synchronises accepted member keys, and exchanges the resources permitted by the configured ACLs. MapSafe can create an authentication group for an account with the required server permission, but this convenience action starts with the current user and does not replace browser-based member invitations or resource-permission administration.

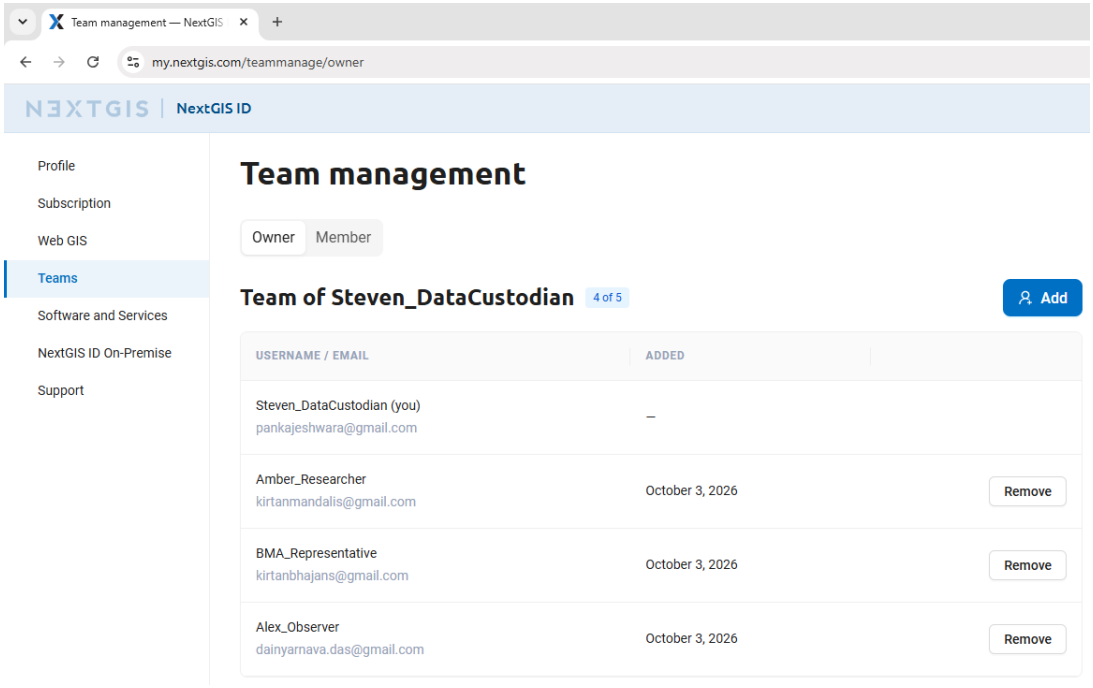


Figure 3.: The NextGIS ID team prepared for the four-account MapSafe Community A evaluation. The displayed names indicate the participants' case-study roles; access to community resources remains governed separately by Web GIS ACLs and OpenPGP recipient selection.

3.2. Encryption identity, key custody, and community public-key exchange

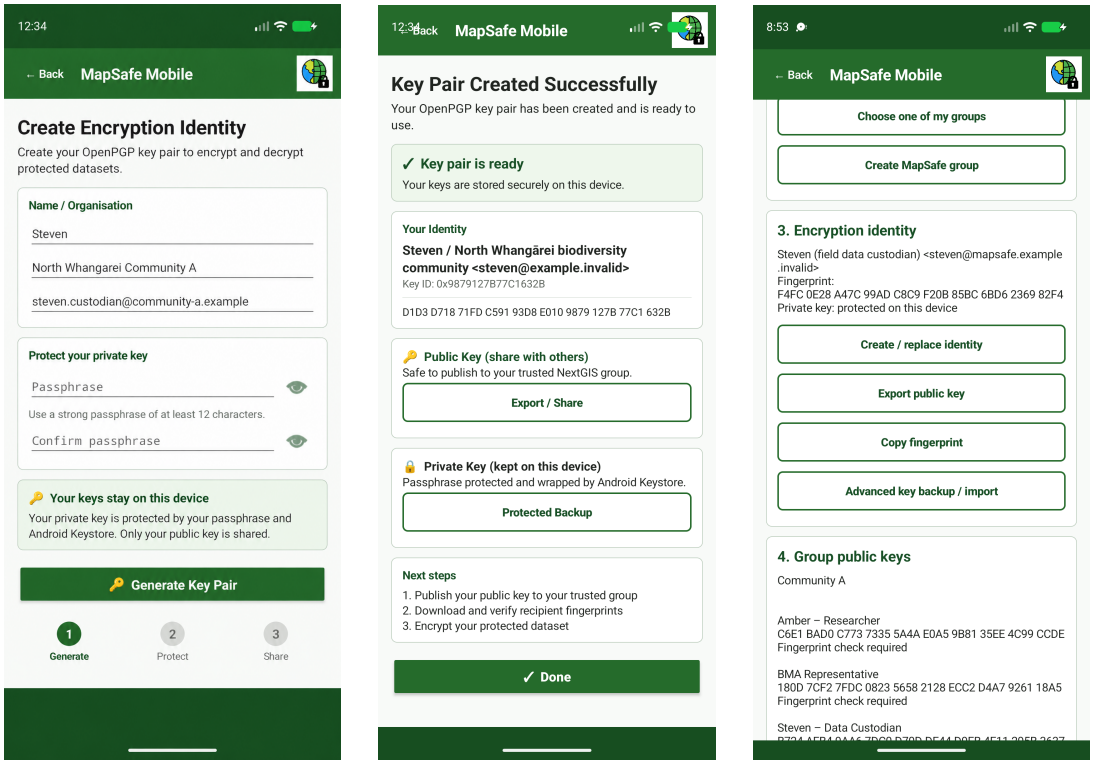
Each participant creates one persistent local OpenPGP identity. Steven supplies a name, organisation, optional email address, and recovery passphrase of at least 12 characters. MapSafe generates an RSA-3072 primary key for certification and signing and a separate RSA-3072 encryption subkey. The transferable secret keyring is protected by an iterated-and-salted SHA-256 string-to-key process, and the application-private copy is additionally encrypted with AES-GCM under a non-exportable *Android Keystore* key. The recovery passphrase is not stored. On later application starts the repository checks the application-private identity store; if a valid identity exists, the encryption screen displays it and does not ask Steven to create another one.

Only public-key material is distributed. A private key is unlocked in memory only for signing or decryption and is never uploaded through the *NextGIS* exchange. Steven, participating BMA representatives, and Amber may each explicitly export their own passphrase-protected secret-key backup before device loss, reset, or application removal. Such backups should be retained on encrypted removable media in a controlled location, with the passphrase held separately by its owner. An end-to-end encrypted channel is preferable if a backup must be transferred remotely. *MapSafe Mobile* cannot recover a forgotten passphrase.

In *Security & Sharing*, Steven selects an authenticated *NextGIS* account and authentication group representing the relevant community. *MapSafe Mobile* creates or uses the group-scoped directory `mapsafe_keys_<group-id>`. Each member publishes a member-owned bucket containing `public-key.asc` and `key-metadata.json`. Pub-

lishing code rejects private-key and passphrase material. The same synchronisation operation is exposed through *Refresh community keys* on the settings and encryption screens.

Discovery is not proof of identity. The client checks current group membership, bucket ownership and uniqueness, manifest/group agreement, agreement between the manifest and downloaded OpenPGP fingerprint, and availability of a usable encryption subkey. The BMA representative's discovered key remains ineligible until Steven compares its full fingerprint through an independent trusted channel and explicitly accepts it. Acceptance pins the fingerprint locally. Changed, missing, duplicate, revoked, or removed-member keys are quarantined and cannot be selected for new packages. Accepted keys are cached for offline use, although membership and key changes require later connectivity. Removing a member prevents selection for future packages but cannot withdraw access to a package already encrypted for that member's key (Wouters et al., 2024).



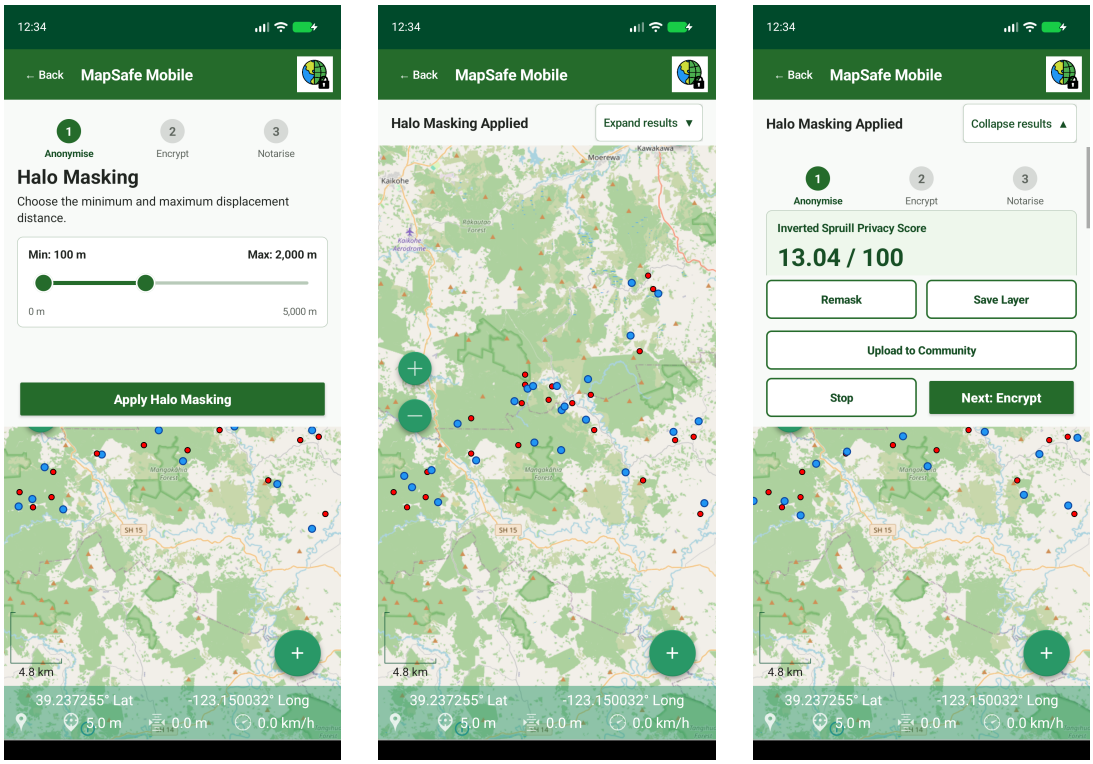
(a) Identity and passphrase. (b) Key ID and fingerprint. (c) Live community key directory.

Figure 4.: Persistent identity and group-scoped public-key exchange. Steven creates a locally protected identity, reviews its fingerprint, and synchronises the published keys of *Amber* and the BMA representative from Community A.

3.3. Geospatial anonymisation and derived-layer sharing

As the first safeguarding step, *MapSafe Mobile* allows a field data custodian to create a less precise representation for people who need to understand the general spatial pattern but are not authorised to receive exact locations. *Steven* begins with the 23-

point *North Whangārei* dataset, chooses *Halo Masking*, and sets the minimum and maximum displacement on one dual-ended range slider (Figure 5a). *MapSafe Mobile* then creates a new masked layer while leaving the authoritative source unchanged. The case-study attributes and one-to-one relationship between records are retained, but each coordinate is replaced by an approximate location that *Steven* can compare with the source on the unobstructed map (Figure 5b) before sharing it with an authorised community recipient.



(a) Dual-ended minimum and maximum displacement control.

(b) Original and halo-masked point layers with the result card collapsed.

(c) Expanded result with the inverted Spruill score and workflow actions.

Figure 5.: MapSafe Mobile halo-masking workflow for the North Whangārei dataset. Steven sets a displacement interval, compares the original and transformed points on an unobstructed map, and expands the result card when the score or next actions are required.

The native Kotlin implementation follows the donut-masking approach demonstrated by *MaskMy.xyz* (Swanlund, Schuurman, and Brussoni, 2020). For each input point $p_i = (\lambda_i, \phi_i)$, a cryptographically secure random generator selects a bearing uniformly from $[0, 2\pi)$ and a displacement within the chosen interval $[r_{\min}, r_{\max}]$. The minimum distance moves the candidate away from its precise source, whereas the maximum distance limits the accompanying loss of spatial utility. Figure 5b overlays the original red points and resulting blue masked points for visual comparison while the collapsed result card leaves most of the mobile map unobstructed.

Protecting privacy generally favours greater displacement, while preserving analytical usefulness favours smaller displacement. To help *Steven* balance these competing aims, *MapSafe Mobile* automatically calculates *Spruill's* disclosure-risk measure after

each masking attempt (Armstrong, Rushton, and Zimmerman, 1999; Swanlund, Schuurman, and Brussoni, 2020). A transformed point contributes to disclosure risk when its corresponding source point remains the nearest member of the original point set. The interface reports the inverted value, P_{inv}, so that a higher score communicates a lower parent-nearest disclosure risk and is easier for a field data guardian to interpret (Figure 5c). The score is guidance rather than a guarantee of anonymity, because disclosure also depends on local geography, attributes, auxiliary information, and the intended use.

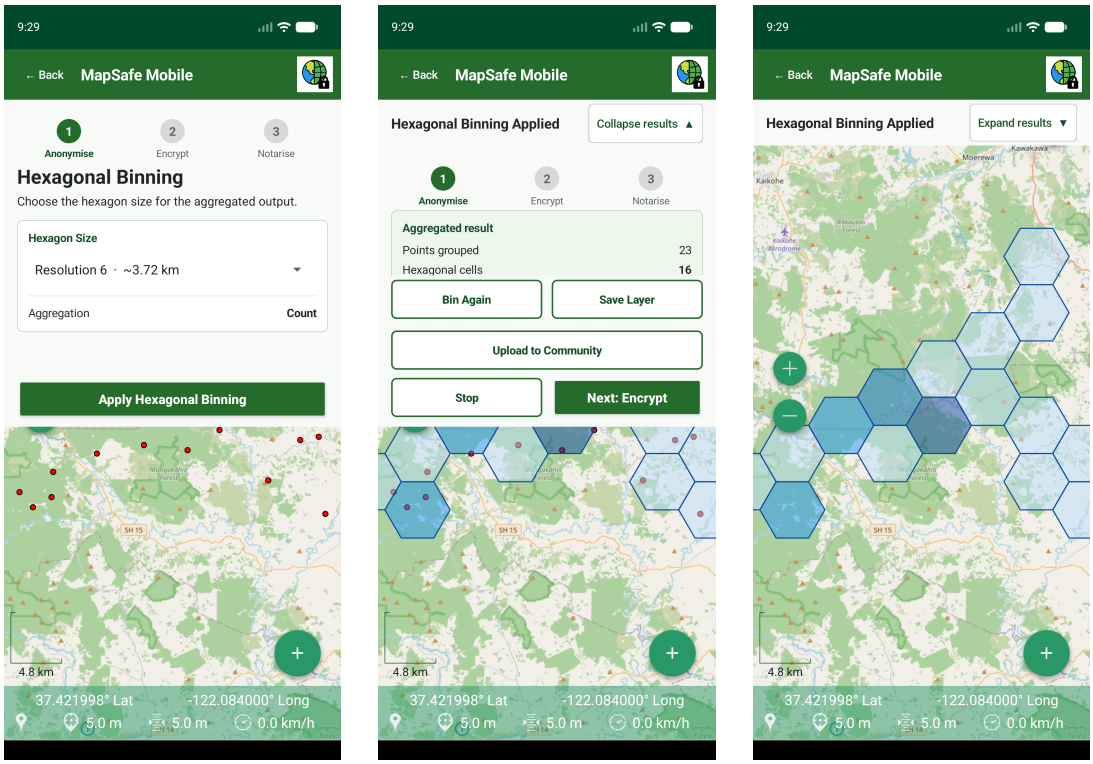
After reviewing the result, *Steven* may collapse the information card to expose almost the entire map, save the layer locally, upload it for an authorised community recipient through the selected *NextGIS* community, stop the safeguard process, or continue to encryption. If the displayed privacy–utility balance is unsuitable, *Remask* returns to the same controls and generates another candidate. Remasking always begins with the precise source rather than the already displaced layer, preventing offsets from being compounded unintentionally.

The physical-device measurements in Table 2 distinguish the fast numerical masking calculation from the complete workflow experienced by the guardian. For the 1,000-point typical dataset, coordinate displacement required a median of 35.72 ms, displacement plus Spruill assessment required 120.73 ms, and the complete production workflow required 5.71 s. The corresponding rich-profile medians were 35.79 ms, 120.87 ms, and 5.96 s. At 2,000 points, the typical and rich complete-workflow medians were 10.81 and 11.52 s respectively. Source reading and output-layer writing, rather than the displacement formula, therefore dominate the mobile operation.

The earlier browser and QGIS studies remain useful contextual comparisons, but their shapefile conversion, nesting, implementation language, hardware, and timing boundaries differ from the present Android experiment. The mobile values are therefore reported from the production NextGIS workflow rather than interpreted as a direct speed-up or slowdown relative to JavaScript in a browser or Python in *QGIS*. This scope distinction also avoids attributing storage and layer-management costs to *Spruill's* nearest-neighbour calculation itself.

Hexagonal Binning is the second anonymisation option in *MapSafe Mobile*. It is intended for situations in which a recipient needs to understand the distribution or concentration of observations but does not require a separate approximate location for every record. Rather than moving each point, MapSafe assigns the original observations to hexagonal cells at a resolution chosen by the data owner. Points sharing the same cell are replaced by one polygon carrying a `point_count` attribute. The resulting layer therefore communicates where observations are concentrated while withholding their exact coordinates. The current interface displays the generated cells as translucent blue polygons; the stored count can also support density-based analysis or styling.

Choosing the resolution requires the same kind of privacy–utility judgement as choosing masking distances. Lower H3 resolutions produce larger cells, grouping observations across a wider area and concealing more positional detail, but they may be too coarse for local analysis. Higher resolutions produce smaller cells and preserve more geographic detail, but can create a sparse surface or cells containing only one observation, which may provide little meaningful protection (Rao et al., 2021). The appropriate level therefore depends on the geographic extent, clustering, sensitivity, and intended use of the dataset rather than on a universal default. In the illustrated *North Whangārei* example, Steven selects resolution 6, which in H3 4.x has an average hexagon edge length of approximately 3.72 km and an average cell area of approximately 36.13 km^2 (H3 Project, 2026). MapSafe groups the 23 observations into 16



(a) Resolution 6 selected for the original points; the displayed 3.2 km text is the legacy H3 3.x estimate.

(b) Twenty-three points grouped into 16 cells.

(c) Collapsed view of the aggregate polygons only.

Figure 6.: MapSafe Mobile hexagonal-binning workflow for the North Whangārei dataset. Steven selects a resolution level, compares the original observations with the occupied cells, and collapses the result card to inspect the anonymised polygon layer without the earlier halo-masked points.

occupied cells, after which Steven compares the source and aggregate layers and inspects the cells without point overlays (Figure 6).

On supported ARM and ARM64 devices, MapSafe performs the indexing with H3 Java 4.1.1 (Brodsky, 2018; Sahr, White, and Kimerling, 2003). The x86/x86_64 emulator used for the screenshots cannot load the packaged H3 native library, so it uses a portable Web Mercator hexagonal grid with resolution-dependent sizes approximating those presented by the interface. Each run creates one binned layer at the selected resolution while leaving the authoritative point layer unchanged. If a different balance is needed for the intended community purpose, *Bin Again* returns Steven to the resolution control and creates another candidate from the precise source. Steven can then save the selected layer, upload it to the chosen NextGIS community, stop the safeguard process, or continue to encryption. The result card can be collapsed so that most of the screen remains available for comparing the aggregation with its geographic context.

MapSafe Mobile supports two levels of data sharing: the original dataset and an anonymised representation. Access is determined by the recipient's role and intended use rather than by fixed categories such as "trusted", "semi-trusted", or "untrusted". A recipient authorised to work with precise locations may receive the original dataset

through multi-recipient *OpenPGP* encryption. A recipient who requires only general spatial patterns receives an anonymised representation created using either halo masking or hexagonal binning. Community membership does not automatically grant access to precise locations; the sovereign data owner remains responsible for selecting the appropriate anonymisation method and parameters, choosing the recipients, and ensuring that each representation is suitable for its intended purpose.

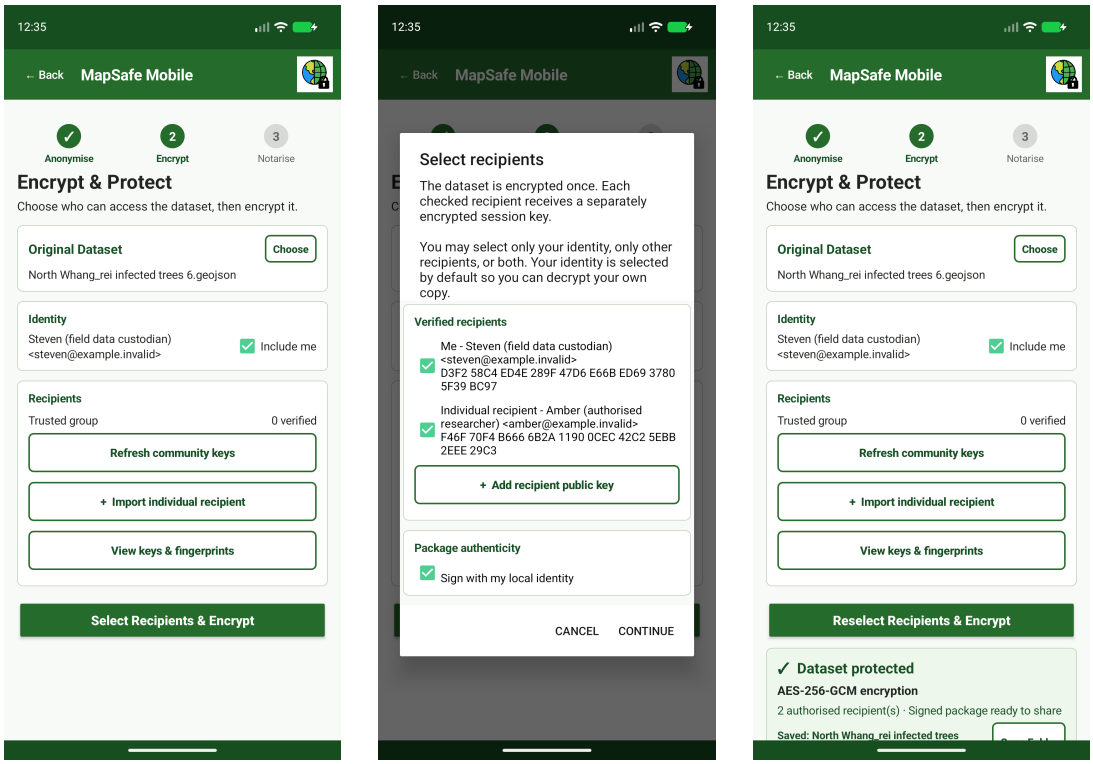
Accepted masked and binned layers can remain on the map, be saved as files, or be uploaded to the selected *NextGIS* community as native vector resources. Local outputs use the fixed public folder `Downloads/MapSafe`; compact feedback shows only the saved filename and an *Open Folder* action. This avoids repeatedly presenting a destination picker in the field. Sharing a derived layer without encryption is an intentional option, but the interface warns that spatial anonymisation reduces risk rather than guaranteeing safety. Attributes, filenames, photographs, timestamps, or small cell counts may still disclose sensitive information (Cassa, Wieland, and Mandl, 2008; Keßler and McKenzie, 2018; Zimmerman and Pavlik, 2008).

3.4. Dataset selection and multi-recipient OpenPGP encryption

Within this two-level model, the original and anonymised representations remain separate artefacts rather than being placed inside a nested, multi-level encrypted volume. In the running example, *Steven* protects the precise *North Whangārei* observations for himself and the authorised BMA representative, while the halo-masked and hexagonally binned layers are released to Amber under separate community permissions. MapSafe encrypts every selected artefact separately, producing one `.pgp` package per dataset; it does not combine several representations inside one package. When several datasets are selected, recipient review is repeated for each package so *Steven* can deliberately apply the same or different recipient set. The *Encrypt* activity can be opened after anonymisation or directly from the Safeguard tab, and its compact *Choose* action lets *Steven* review or change the current dataset before recipient selection (Figure 7a).

Steven next chooses exactly who should be able to recover the package. If his persistent local identity is available, *MapSafe* selects *Steven's* public key as a self-recipient by default so that he can decrypt the package later. For the tested release, Steven retains that selection and adds the independently verified public key of the BMA representative (Figure 7b); Amber is deliberately not selected for the encrypted original. The same mechanism can address other authorised recipients in a different release, but recipient selection remains explicit for every package. If *Steven* must create an encryption identity at this point, MapSafe returns to recipient selection afterwards rather than beginning encryption automatically.

After the recipients have been confirmed, *MapSafe* uses standard *OpenPGP* hybrid encryption to create one `.pgp` package. The application compresses the selected dataset, generates a fresh random AES-256 content-encryption key, and encrypts the dataset once using the RFC 9580 version-2 Symmetrically Encrypted Integrity Protected Data profile and AES-256-GCM (Wouters et al., 2024). It then encrypts that same content key separately with each selected recipient's RSA encryption subkey. These public-key encrypted session-key envelopes and the encrypted payload are stored together inside the single package. Adding another recipient therefore adds a small key envelope rather than another encrypted copy of the dataset. The same accepted community members are assigned the package's NextGIS ACL, which controls discovery and download, while OpenPGP controls decryption.



(a) Selected North Whangārei dataset and persistent local identity.

(b) Selection of self and authorised community recipients.

(c) Protected package saved with continuation actions.

Figure 7.: MapSafe Mobile multi-recipient encryption workflow. Each selected dataset is encrypted once, which is independently wrapped for every selected OpenPGP public key. The same accepted recipients define the default NextGIS access permissions for the uploaded encrypted package, while OpenPGP controls decryption. If several datasets are selected, each becomes a separate protected package.

This design removes the need to distribute a shared symmetric key or a common dataset passphrase. Each recipient uses their own private key to recover their copy of the session key, while their personal passphrase is used only to unlock that private key locally. Neither the private key nor its passphrase is added to the package or uploaded to NextGIS. If signing is enabled, the owner's passphrase temporarily unlocks the local signing key and MapSafe embeds a SHA-256 OpenPGP signature, allowing recipients to distinguish confidentiality from evidence about the sender. New packages use the AES-256-GCM profile, while decryption remains compatible with the earlier AES-256/OpenPGP-MDC MapSafe profile.

Table 2 is the mobile equivalent of the masking, encryption, and decryption tables reported for the earlier browser and QGIS versions of MapSafe. The primary datasets contain 50, 250, 500, and 1,000 points because these sizes reflect observations likely to be collected during a NextGIS Mobile field session; 2,000-point datasets provide an additional stress case. Every point-count class has two GeoJSON attribute profiles. Both contain the same 18-field typed schema, but the *rich* profile uses longer, non-repeating notes and stewardship text and is approximately four times larger than the corresponding *typical* file. Attachments are excluded because the current MapSafe

Table 2.: Median MapSafe Mobile processing times on the Samsung SM-A145F after one warm-up and five measured repetitions.

Points	Profile	Input	Masking			OpenPGP		Output
		KiB	Core (ms)	Core+SM (ms)	Complete (s)	Encrypt (s)	Decrypt (s)	PGP KiB
50	Typical	39.76	1.90	5.35	0.32	2.34	2.32	15.12
50	Rich	163.20	1.81	4.11	0.32	2.42	2.38	111.34
250	Typical	198.98	12.61	78.33	1.75	2.41	2.36	67.74
250	Rich	816.16	8.97	21.40	1.43	2.80	2.63	550.12
500	Typical	398.12	17.61	46.76	2.87	2.49	2.42	133.53
500	Rich	1,632.49	16.35	41.99	2.79	3.36	2.93	1,098.60
1,000†	Typical	796.39	35.72	120.73	5.71	2.68	2.52	265.18
1,000	Rich	3,265.13	35.79	120.87	5.96	4.24	3.52	2,195.48
2,000	Typical	1,594.87	65.04	180.31	10.81	3.06	2.73	528.41
2,000	Rich	6,532.37	70.06	200.23	11.52	6.14	4.69	4,389.20

GeoJSON exporter does not embed NextGIS attachments.

The performance experiment was conducted on a physical Samsung Galaxy A14 (SM-A145F; product `a14nsxx`) equipped with a Samsung Exynos 850 system-on-chip, eight logical processors, and approximately 3.60 GiB of usable memory (4 GB class). Ten synthetic GeoJSON datasets represented five point counts—50, 250, 500, 1,000, and 2,000—under typical and rich attribute profiles. For every dataset, the test harness evaluated the complete halo-masking workflow, signed OpenPGP encryption, and verified OpenPGP decryption, producing 30 dataset–operation cells. Their execution order was randomised using the fixed seed 20260811. Each cell was completed once as an unreported warm-up and then five times for measurement. Table 2 reports the median of these five measurements to two decimal places; the corresponding first and third quartiles are retained in the archived result files. Halo masking used fixed 100–2,000 m displacement bounds. The OpenPGP tests used one RSA-3072 identity generated before measurement, one selected recipient, signing enabled, compression, AES-256-GCM encryption, integrity checking, and signature verification.

The three masking columns deliberately use different boundaries. *Mask core* records source-coordinate normalisation, cryptographically secure displacement, and spherical destination calculation. *Core+SM* includes Spruill's nearest-neighbour assessment on that same candidate. *Complete mask* begins when the already selected NextGIS source layer is read and ends after output-feature construction, transactional insertion, spatial-index rebuilding, and map saving. The cryptographic cells exercise the same production OpenPGP engine over in-memory byte streams. Encryption starts with prepared WGS84 GeoJSON bytes and ends when the signed single-recipient package bytes have been produced; decryption includes private-key unlocking, payload recovery, integrity checking, and signature verification and ends when the plaintext bytes have been recovered. Android file-picker interaction and filesystem input/output are not part of these two cryptographic timings. Human interaction, initial dataset import, UI rendering, community transfer, blockchain access, and result-log writing are also excluded. Hexagonal binning was not timed in this experiment and must be evaluated separately on the physical H3 implementation.

Each timing is reported to two decimal places. Milliseconds are retained for the two numerical masking phases because expressing them to hundredths of a second would

obscure their differences; complete masking, encryption, and decryption are shown in seconds. Encryption and decryption concern one dataset in one single-recipient package rather than nested representation levels, and the 2,000-point rows are stress cases. Because every cryptographic run used one recipient, the results do not quantify the time or package-size overhead of adding recipients. The 1,000-point typical masking median marked † comes from a verification repeat after transient device contention affected three repetitions in the complete run; its cryptographic medians remain from the complete run.

Only one encryption and one decryption column are required because *MapSafe Mobile* does not repeatedly encrypt nested fine, medium, and coarse representations. The anonymised output remains a separate shareable layer, while the original dataset is encrypted once for the selected recipients. For 50 points, the typical and rich profiles each completed the masking workflow in a median of 0.32 s. At 1,000 points, the typical profile required 5.71 s and the rich profile 5.96 s. The similar values at a fixed point count, despite the rich input being approximately four times larger, show that feature count and *NextGIS* layer I/O influence complete masking more strongly than exported GeoJSON byte size. For the 1,000-point rich dataset, coordinate displacement itself required a median of 35.79 ms and displacement plus Spruill assessment required 120.87 ms. These calculation times are small compared with the 5.96 s complete workflow because reading, constructing, inserting, indexing, and saving the derived layer dominate the operation experienced by the guardian.

Signed encryption and verified decryption showed a fixed cryptographic and stream-processing cost for small inputs and then increased with GeoJSON byte size. The 50-point typical dataset required median times of 2.34 s for encryption and 2.32 s for decryption. At 1,000 points, the typical dataset required 2.68 s and 2.52 s respectively, whereas the larger rich dataset required 4.24 s and 3.52 s. The 2,000-point rich stress file (6.38 MiB) required 6.14 s for encryption and 4.69 s for decryption, corresponding to median throughputs of 1.04 and 1.36 MiB/s. The 1,000-point rich dataset reached 0.75 MiB/s for encryption and 0.91 MiB/s for decryption. Compression made every `.pgp` package smaller than its GeoJSON input. The archived quartiles show generally modest variability, but they are omitted from the compact table in favour of one median per cell. Repeating the initially contended 1,000-point typical masking cell produced a complete-workflow median of 5.71 s, which is the masking value reported in Table 2.

These *Android* values should not be compared directly with the throughput reported for the earlier browser and QGIS implementations. The mobile experiment includes the production NextGIS read/write path and signed OpenPGP processing, uses different input formats and hardware, and tests field-scale rather than national datasets. The 1,000-point rich dataset provides a useful upper reference for the intended field workflow: its median complete-masking, encryption, and decryption times were 5.96, 4.24, and 3.52 s respectively. The longest measured median was 11.52 s for complete masking of the 2,000-point rich stress dataset. Raw measurements, phase timings, dataset hashes, device metadata, and both the complete and verification runs are archived with the OSF package (https://osf.io/cyfu7).

The completed package is written automatically to `Downloads/MapSafe`, and the result card confirms the saved filename (Figure 7c). The owner may stop at this point, continue to notarisation, upload the package to the selected NextGIS community, or reselect recipients and create a new package. The package payload remains opaque to *NextGIS* because the service does not possess an addressed private key; the package filename and descriptive registry metadata remain visible to authorised community

members and should therefore avoid sensitive wording. Encryption and decryption performance for the intended 50–1,000-point mobile datasets is reported in Table 2; unlike the earlier nested scheme, the measured processing time increases principally with dataset size.

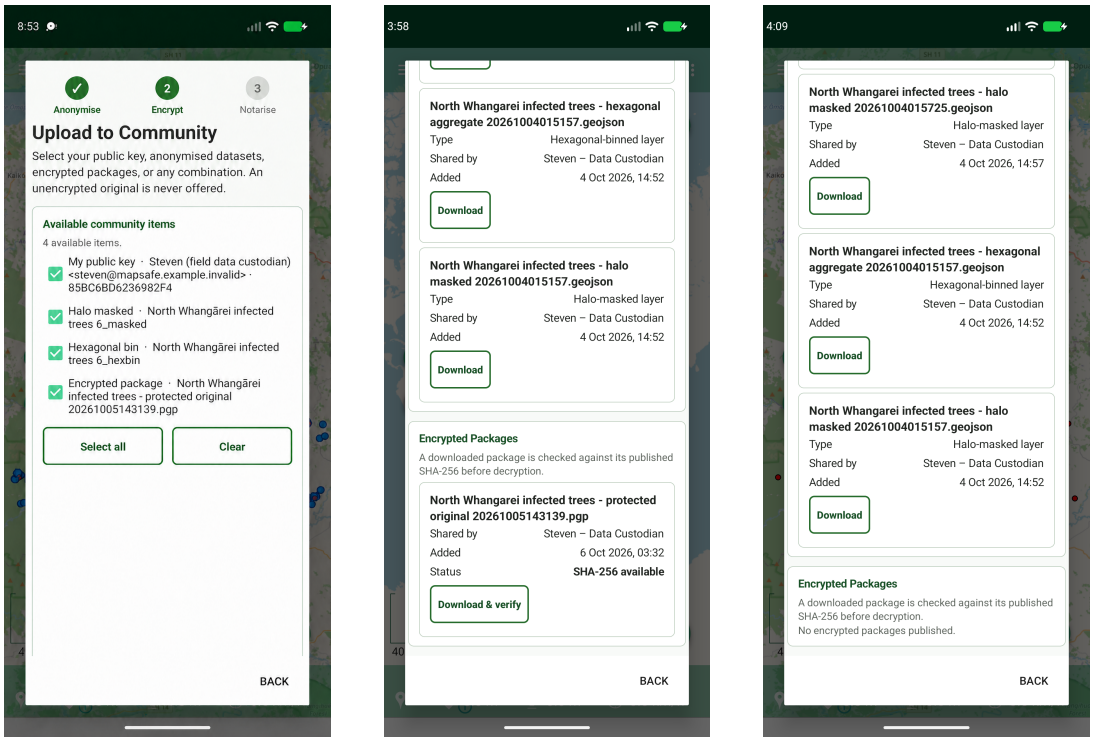
3.5. NextGIS community resources and protected-resource exchange

MapSafe uses different *NextGIS* resources for different artefacts. Public keys use a dedicated group-scoped key directory. Halo-masked and hexagonally binned results are uploaded as native vector resources so authorised members can view or reuse them through the Web GIS. Encrypted `.pgp` packages remain opaque attachments associated with a publisher-owned package registry. A registry entry records the package filename, SHA-256 digest, publisher and group identifiers, creation time, status, and available blockchain metadata. *MapSafe* never offers an unencrypted original dataset through this upload workflow.

Steven opens *Upload to Community* from the Safeguard tab and may select a public key, one or more anonymised datasets, one or more encrypted packages, or any combination of these items (Figure 8a). Each selection is published under his authenticated account and currently selected group. On the BMA representative's device, *Community Packages* lists the recipient-restricted protected original together with any anonymised layers permitted by its ACL (Figure 8b). Amber sees the community's halo-masked and hexagonally aggregated datasets but is not offered the encrypted original because her key was not selected for that package (Figure 8c). Public keys can be synchronised for later recipient selection, anonymised layers can be downloaded as GeoJSON, and authorised protected packages can be downloaded to `Downloads/MapSafe` for verification and decryption.

A four-account Premium acceptance test was executed against the live North Whangārei community. It published *Steven's*, *Amber's*, and the *BMA representative's* public-key records, two anonymised layers, and one original dataset encrypted and signed for Steven and the BMA representative. As shown in Figure 9, these resources are organised within separate *Public Keys*, *Anonymised Datasets*, and *Encrypted Packages* groups under MapSafe Community A. Steven and the BMA representative downloaded byte-identical copies of the protected package; the BMA representative recovered a byte-identical copy of the original and verified Steven's signature. Amber could list and download the two anonymised layers but could neither list nor directly read the protected package, while the external test user could not discover or directly read any Community A resource. Wrong-key and wrong-passphrase attempts produced no plaintext.

When an encrypted community package is selected, *MapSafe* downloads it, recalculates its SHA-256 digest, and compares the result with the digest recorded in the NextGIS registry before passing the package to the integrity screen; a mismatch is rejected. Access to the resource hierarchy shown in Figure 9 remains subject to the Web GIS ACLs configured by the administrator. Group membership identifies the relevant community, but MapSafe does not infer that every authenticated NextGIS user may read every resource; access is determined separately for public keys, anonymised datasets, and recipient-restricted encrypted packages.



(a) Steven selects items to publish.

(b) The BMA representative receives the protected original.

(c) Amber receives anonymised layers only.

Figure 8.: Observed Premium community exchange for the North Whangārei case. Steven publishes selected resources; the BMA representative receives the recipient-encrypted original, whereas Amber's view contains only the authorised anonymised layers.

3.6. Notarisation and blockchain-verification boundary

The *Notarise* screen automatically receives the encrypted package created by the preceding step and calculates its SHA-256 digest. If *Steven* opens it independently, he can select an existing encrypted package. Only the active network name is shown on the operational screen. Network details are configured in *Security & Sharing*, where *MapSafe* stores encrypted app-private profiles for Ethereum Sepolia, Ethereum Mainnet, or a custom EVM-compatible network. Preflight checks validate the RPC chain identifier, deployed bytecode, expected `mintNFT(string)` and `locations(uint256)` selectors, and ERC-721 interface support before the profile is used.

For a new notarisation, *MapSafe* normalises the encrypted package basename and appends an underscore followed by its 64-character lowercase SHA-256 digest. This filename–digest string is encoded as the argument to `mintNFT(string)`. Binding the public package name to the digest gives the resulting assertion more evidential context than a hash alone: the signing wallet asserted that the named encrypted package had those bytes at the transaction time. It does not expose the plaintext dataset or cryptographic secrets, and it cannot by itself prove the human identity controlling the wallet, prevent another transaction from recording a competing assertion, or make a user-selected filename an intrinsic property of the bytes. Neutral encrypted-package names should therefore be used where a descriptive name would itself disclose sensitive

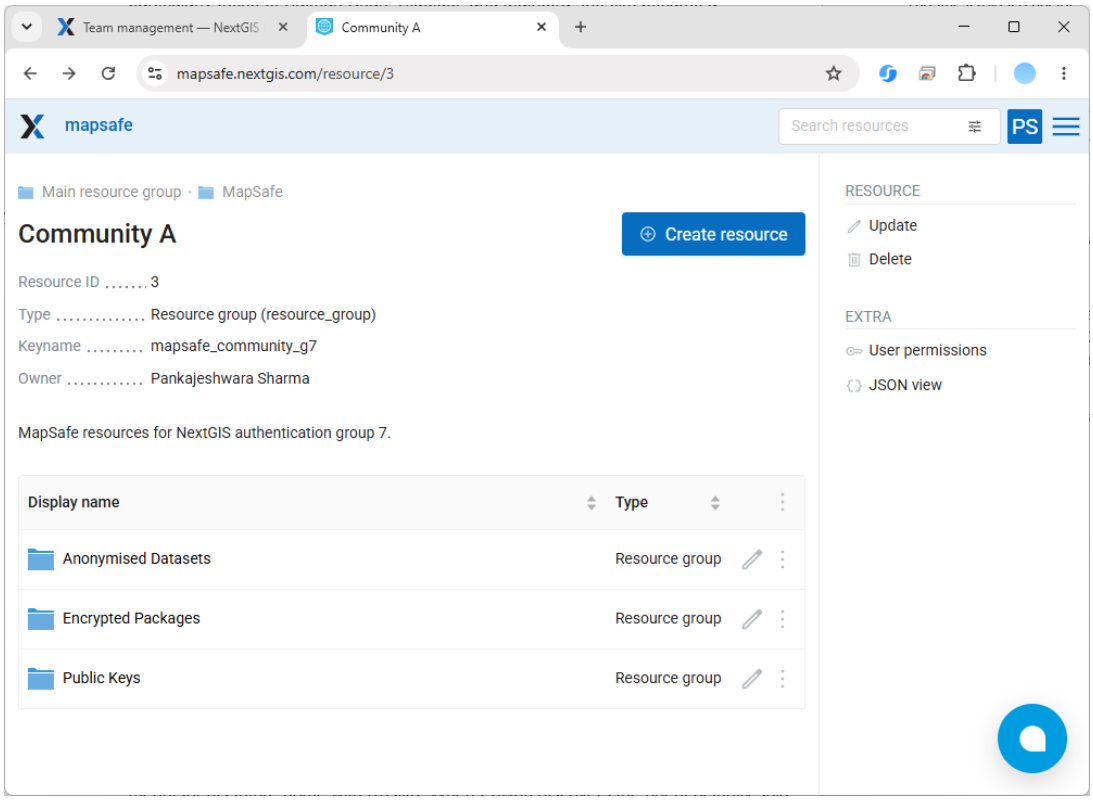
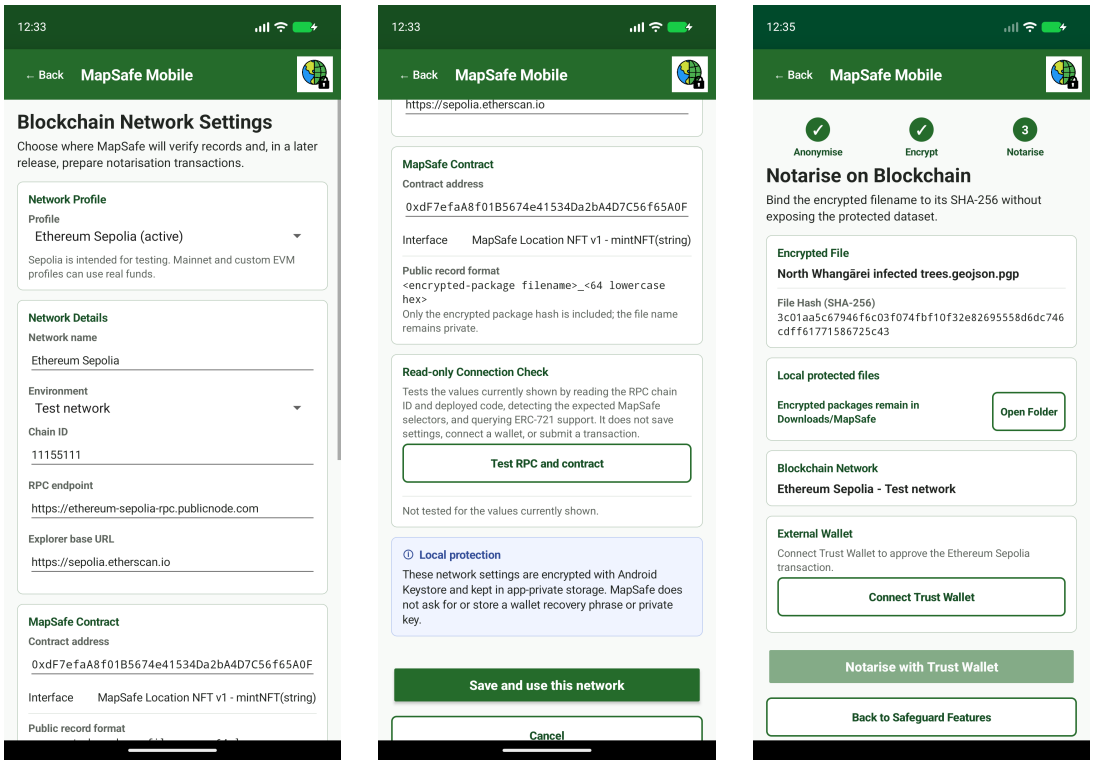


Figure 9.: The MapSafe Community A resource structure in NextGIS Web. Separate resource groups organise community members' OpenPGP public keys, anonymised datasets, and encrypted packages. Access to these resources is governed by Web GIS ACLs associated with the corresponding NextGIS authentication group.

information.

MapSafe constructs a zero-value transaction and hands it through Reown WalletConnect to an installed external wallet, such as *Trust Wallet* or *MetaMask*, for explicit review, fee approval, and signing (Reown, 2026). The application neither collects nor stores the blockchain private key or recovery phrase. After submission, *MapSafe* retrieves the receipt, verifies that the call was mined successfully on the configured chain and contract, and compares the decoded filename and digest with the local package before reporting success. The transaction hash and network can then be associated with the corresponding community package. This function is complementary to *OpenPGP*: the blockchain record supplies an independently retrievable public assertion about the protected package, while recipient encryption and the optional OpenPGP signature establish confidentiality, package integrity, and signer status.

Network and contract details, including RPC and contract preflight checks, remain in *Security & Sharing*; the operational Notarise screen carries forward the encrypted filename, calculates its digest, displays the active network and connected wallet, and requests external-wallet approval. Figure 10 shows the available configuration, contract-checking, and wallet-handoff screens. The Verification screen shown in Figure 11b also accepts a transaction reference and, when one is supplied, performs the local-to-chain filename and digest comparison.



(a) Active EVM network profile.

(b) Contract and RPC preflight controls.

(c) Filename–digest record and wallet hand-off.

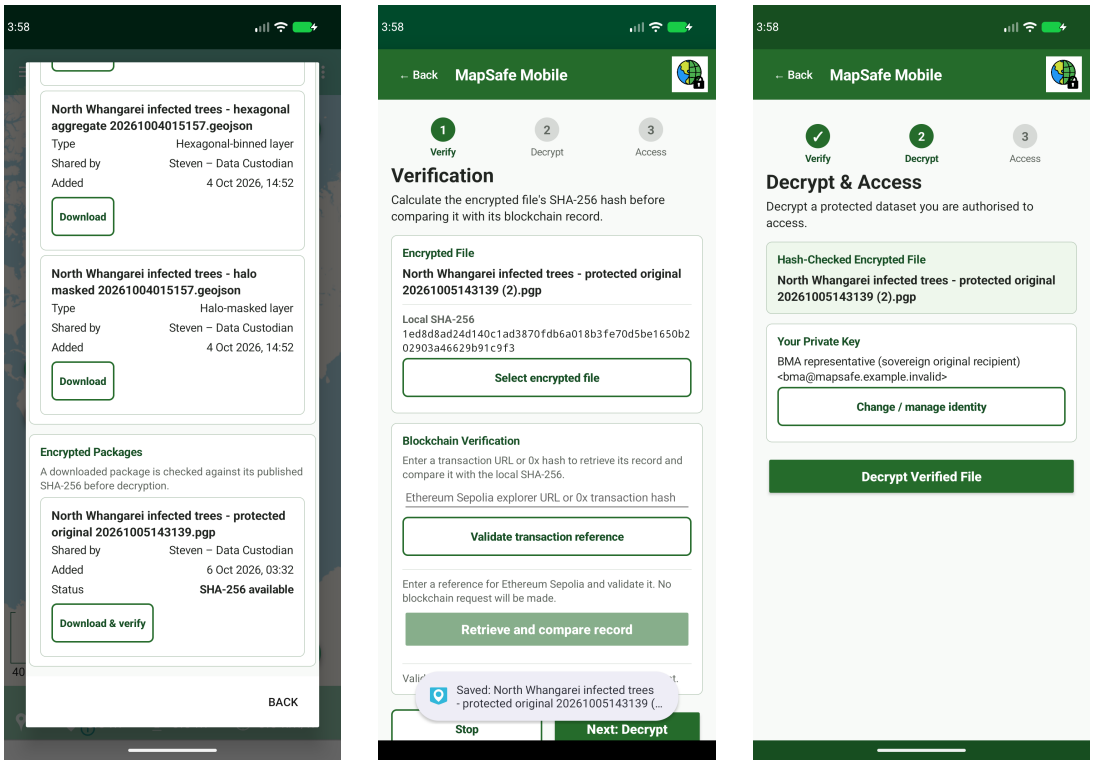
Figure 10.: MapSafe Mobile's notarisation workflow. Steven selects an EVM network profile, validates the configured RPC endpoint and contract, and prepares a filename–SHA-256 assertion for approval and signing in an external wallet. These screens document configuration and transaction preparation; external-wallet approval and mined-transaction confirmation occur subsequently.

3.7. Verification, decryption, and map display

Access begins either from the community package list or the Android file picker. In the tested release, the BMA representative opens *Community Packages* and can discover the encrypted original because the package registry ACL includes that account; Amber can discover only the authorised anonymised layers. Selecting *Download & verify* saves the `.pgp` file to `Downloads/MapSafe`, calculates SHA-256, and compares the downloaded bytes with the digest recorded in the NextGIS registry. A mismatch is rejected before the package reaches the decryption stage.

The Verification screen can additionally accept a compatible blockchain transaction URL or hash. When one is supplied, MapSafe retrieves the transaction and receipt, checks the configured chain, contract, call, and receipt status, strictly decodes the record, and compares both the normalised encrypted filename and digest. Earlier hash-only MapSafe records remain readable but establish only a digest match. This blockchain comparison is optional for a package that was not notarised; the community-package evidence in Figure 11b therefore shows the completed local and registry check before decryption, while the optional transaction field remains available.

The package contains one encrypted dataset. MapSafe inspects its OpenPGP recipient identifiers and locates the BMA representative's matching local secret encryption



(a) The BMA representative discovers the protected original.

(b) Downloaded bytes are hashed and checked against the community record.

(c) MapSafe locates the addressed BMA private key.

Figure 11.: Recipient-side access to the protected original. The BMA representative discovers and downloads the authorised community package, verifies its recorded digest, unlocks the matching local OpenPGP private key, and would later display the recovered North Whangārei dataset after the earlier map layers are cleared.

key (Figure 11c). The representative's personal passphrase unlocks that private key in memory, the key recovers the AES session key from its recipient envelope, and the session key decrypts the payload. The session key is therefore an internal cryptographic value rather than a second file that the recipient must manage. A package addressed only to another recipient is rejected, as confirmed by the wrong-key test.

AEAD/MDC integrity and signature authenticity are evaluated separately. Modified or structurally invalid packages are rejected before plaintext is accepted, while a present signature is reported as valid, invalid, or made by an unknown signer. In the four-account acceptance test, the BMA representative recovered a byte-identical GeoJSON copy and MapSafe verified Steven's signature. The result is saved to `Downloads/MapSafe`, imported as one local NextGIS vector layer, selected, and zoomed to its extent. Previously loaded layers are cleared before display, so the access workflow proceeds directly to the recovered North Whangārei original rather than opening an intermediate dataset catalogue (Figure 11d).

4. Usability-Informed Design and Evaluation

The principal usability objective of *MapSafe Mobile* is to make safeguarding and access decisions understandable during field collection. The interface separates *Safeguard* and *Access*, shows progress and consistent *Back*, *Stop*, and *Next* actions, keeps masking and binning results visible through collapsible cards, recognises existing identities and community keys, and displays the recovered layer directly after decryption. These choices reduce repeated navigation while keeping the consequences of each operation visible.

The evaluation approach was adapted from the earlier *MapSafe QGIS plugin* study (Sharma, Young, and Dubrava, 2025) and follows the effectiveness, efficiency, and satisfaction dimensions of ISO 9241-11:2018 (ISO, 2018). The evidence is deliberately distinguished: effectiveness was tested through the mobile workflow, efficiency was assessed using physical-device measurements, and the earlier interface findings were used as formative design evidence. Mobile-user satisfaction was not measured.

For *effectiveness*, the first author completed the 12 safeguarding and access tasks listed in Appendix Table 4, twice from the same prepared North Whangārei and Community A state. All 24 trials were completed successfully, giving a 100% task-completion rate. No user or system errors, critical errors, assistance events, unintended plaintext uploads, incorrect-recipient encryptions, false integrity acceptances, or unrecovered datasets were recorded. The authoritative layer remained unchanged, derived layers were created separately, and the community workflow did not offer an unencrypted original for publication. This demonstrates functional effectiveness under prepared conditions, not independent first-time-user usability.

For *efficiency*, the physical-device measurements in Table 2 show that the principal local operations are practical for field-scale datasets. The 1,000-point rich dataset required median times of 5.96 s for complete masking, 4.24 s for signed encryption, and 3.52 s for verified decryption. Notarisation was excluded from the fixed processing time because wallet approval, transaction submission, connectivity, and blockchain confirmation are external and variable.

Formative interface evidence came from the QGIS study's heuristic inspection and comparable safeguarding and access tasks (Nielsen, 1995). Its recommendations concerning status visibility, navigation, file selection, recognition rather than recall, passphrase handling, screen density, output management, help, and error recovery were mapped to the mobile activities; the resulting traceability audit is provided in Appendix Table 3. Progress and validation feedback were implemented directly, while large desktop panels were adapted as collapsible mobile cards and file selection was followed by format and content validation. These inherited findings are design evidence, not satisfaction results from mobile participants; independent testing of mobile satisfaction, accessibility, and learnability remains necessary.

5. Discussion

Protecting sensitive geospatial information in a mobile GIS is not solely a technical problem; it is also a governance problem concerning ownership, spatial detail, access, and subsequent use. Distributed spatial-data research emphasises the need for explicit ownership and access-control protocols when data are produced and shared across multiple actors (Hojati et al., 2023), while group-privacy research shows that disclosure can harm communities through contextual inference, aggregation, and the

combination of datasets (Masinde et al., 2023). Blockchain research identifies integrity preservation and traceable sharing as useful applications, but not as substitutes for confidentiality, legitimate authority, or access control (Zhao et al., 2022). MapSafe Mobile operationalises these distinctions by separating masking and aggregation, OpenPGP recipient control, NextGIS community permissions, and optional blockchain anchoring. Its contribution is therefore not a new anonymisation or cryptographic primitive, but an integrated mobile workflow that applies established safeguards before synchronisation while leaving representation, recipient selection, and community permissions under the data custodian's control.

The two-level model keeps the authoritative original separate from an anonymised representation. Halo masking and hexagonal binning determine what geographic detail is disclosed, while OpenPGP determines who can recover a selected dataset. These are independent decisions: a dependable community member may have no legitimate reason to receive precise coordinates, whereas an authorised BMA representative may need them for sovereign oversight. This distinction makes the release decision explicit for Steven, Amber, and the BMA representative rather than reducing their roles to a single trust label. Unlike the earlier nested MapSafe scheme, each selected dataset is protected as its own artefact and, when encrypted, as a separate `.pgp` package. This lets Steven select recipients for each representation while keeping protection within the established field-data environment.

In terms of *performance analysis*, the physical-device results indicate that the intended field-scale operations are practical on a modest Android handset. For the 1,000-point rich dataset, complete halo masking, signed OpenPGP encryption, and verified decryption required median times of 5.96, 4.24, and 3.52 s; the 2,000-point stress case required 11.52, 6.14, and 4.69 s. Masking time was driven mainly by feature count and the reading, construction, indexing, and saving of the NextGIS layer, whereas cryptographic time increased with exported bytes, particularly for rich attributes. Compression made the resulting encrypted packages smaller than their GeoJSON inputs. Network transfer, wallet interaction, blockchain retrieval, and transaction confirmation were not included because they depend on external conditions.

As for *security analysis*, we found that local processing reduces unnecessary disclosure to remote services, but it does not remove the need to trust the handset: original coordinates exist in memory and local storage before protection. *Android Keystore* protection, passphrase-protected keys, device encryption, secure updates, controlled backups, and incident response are therefore complementary requirements. The same boundary applies to retained attachments, screenshots, metadata, and automatic device backups. Masking and aggregation improve the privacy–utility trade-off but do not guarantee anonymity; clustering, attributes, timestamps, photographs, external datasets, singleton cells, and sparse H3 resolutions may still support re-identification (Cassa, Wieland, and Mandl, 2008; Keßler and McKenzie, 2018; Zhang and McKenzie, 2022; Zimmerman and Pavlik, 2008).

OpenPGP provides recipient-specific confidentiality by encrypting each selected dataset once with a fresh AES-256 session key and wrapping that key for every selected recipient. In the tested release, Steven encrypted and signed the original for himself and the BMA representative, while Amber received only the authorised masked and binned layers. Fingerprint review is still necessary because a public key is not proof of identity; membership changes or compromised keys may also remain hidden in an offline cache until synchronisation resumes (Wouters et al., 2024). The Premium test illustrates the separation of controls: NextGIS ACLs governed discovery and download, OpenPGP recipients governed decryption, and wrong-key or wrong-passphrase

attempts produced no plaintext. NextGIS therefore remains trusted for administration, availability, ACL enforcement, and key distribution, but it does not hold the private keys (NextGIS, 2026b; Wouters et al., 2024).

Blockchain notarisation has a narrower role than confidentiality or authority. MapSafe records a normalised encrypted-package filename with its SHA-256 digest so that a verifier can establish byte identity for the named package. This does not prove that the observations are accurate, that collection was authorised, or that the wallet controller had legitimate authority. The filename must therefore avoid sensitive information, and production deployment still requires contract review, network and fee policies, wallet recovery, and failure handling.

In terms of *usability analysis*, *MapSafe Mobile* was designed to make secure behaviour a continuation of the field workflow. Progress indicators, consistent *Back*, *Stop*, and *Next* actions, early layer validation, identity recognition, deliberate recipient selection, collapsible result cards, and predictable output storage reduce technical barriers while keeping the consequences of protection visible. Carrying the selected dataset between stages, recognising stored identities, listing community artefacts, and displaying the recovered layer directly also reduce repeated file navigation. In the first-author effectiveness evaluation, the 12 safeguarding and access tasks were each completed twice (24 trials) without user or system errors, incorrect-recipient encryptions, false integrity acceptances, unintended plaintext uploads, or unrecovered datasets. This demonstrates functional effectiveness under prepared conditions, not independent first-time-user usability.

The design was also informed by the earlier evaluation of the *MapSafe QGIS plugin* (Sharma, Young, and Dubrava, 2025), which examined effectiveness, efficiency, and satisfaction using ISO 9241-11:2018 (ISO, 2018) and Nielsen's heuristics (Nielsen, 1995). The mobile implementation adapted its recommendations on status visibility, navigation, file selection, recognition rather than recall, passphrase handling, screen density, output management, help, and error recovery. Desktop panels became collapsible cards, passphrase visibility was added through eye controls, and existing identities and community keys are recognised automatically. An independent satisfaction and accessibility study remains necessary because the earlier participants did not evaluate the Android application.

Our work is not without *limitations and recommendations for future work*. Our evidence is bounded by one Samsung Galaxy A14, one recipient per encrypted package, and local timings that excluded network, wallet, and blockchain latency. The effectiveness assessment was conducted by the first author and did not measure independent satisfaction, accessibility, learnability, or retention. Future benchmarking should vary recipient counts, devices, Android versions, and connectivity, including offline reconnection and confirmation delays.

Deployment also needs stronger lifecycle controls. The current implementation supports one local OpenPGP identity per installation but not complete organisational certification, rotation, revocation, or managed recovery. Backup and restoration should be tested across device replacement, application removal, compromised accounts, membership and ACL changes, stale caches, and service interruption, with interoperability checked against independent OpenPGP tools (NextGIS, 2026b; Wouters et al., 2024). Production notarisation also requires governed contract deployment, wallet recovery, fee and network policies, confirmation handling, and failure recovery. Future releases should address lines, polygons, trajectories, rasters, attachments, and metadata, while testing context-aware masking, organisation-defined H3 rules, and minimum-count warnings. Independent usability, interoperability, and participatory studies with the

relevant BMAs are needed to determine acceptable spatial detail, recipient authority, retention, reuse, terminology, and accountability. MapSafe Mobile provides mechanisms for those decisions; it cannot confer sovereignty or cultural legitimacy by itself (Carroll et al., 2020; Jennings et al., 2023; Kukutai and Taylor, 2016; Lovett et al., 2019; Reyes-García et al., 2022).

6. Conclusion

The growing use of mobile GIS for field data collection highlights the need to protect sensitive geospatial information at the point where it is created. *MapSafe Mobile* addresses this challenge by integrating halo masking, hexagonal binning, encryption, community sharing, blockchain-based notarisation, verification, and decryption into NextGIS Mobile. Bringing these functions into the field-data environment allows custodians to decide what spatial detail should be disclosed, who should receive it, and how its integrity can be verified before the original observations are synchronised or transferred elsewhere. This approach advances the concept of *capture-time sovereignty*, in which safeguarding becomes part of data collection rather than an activity postponed until the data reach a desktop computer or external service.

Protecting a dataset introduces an additional step into the mobile GIS workflow, but failing to safeguard precise locations can lead to irreversible disclosure, loss of trust, and diminished community control. *MapSafe Mobile* demonstrates how anonymisation, recipient-based encryption, resource permissions, and integrity evidence can be combined within an accessible workflow without replacing the familiar map-based environment. Its separation of original and anonymised representations enables disclosure to be matched to the recipient's authorised purpose rather than to a fixed hierarchy of trust. Local processing also reduces reliance on third-party services for handling plaintext, while NextGIS community resources support controlled collaboration between custodians, researchers, sovereign parties, and other authorised recipients. As an open-source implementation, *MapSafe Mobile* additionally provides a transparent and adaptable foundation for organisations seeking to incorporate geoprivacy into mobile field practice.

MapSafe Mobile does not by itself guarantee anonymity, establish legitimate authority, or confer Indigenous Data Sovereignty. These outcomes depend on governance, identity verification, appropriate anonymisation parameters, key management, recipient responsibilities, and continuing community oversight. Future work should extend the approach to additional geospatial data types, strengthen key recovery and revocation, evaluate alternative masking strategies, assess production blockchain deployments, and conduct independent and participatory evaluations with the communities whose information is represented. By embedding geoprivacy within mobile GIS, *MapSafe Mobile* demonstrates a practical direction for protecting sensitive observations before external synchronisation and for making disclosure and access decisions more visible, accountable, and closely connected to the people responsible for the data.

Appendix

Evaluation protocol and interpretation. The effectiveness evaluation was conducted by the first author, who is familiar with GIS, Android field mapping, and the MapSafe implementation. The evaluator used the prepared North Whangārei dataset

and configured Community A state, and completed the 12 safeguarding and access tasks in the order shown in Table 4. Each task was performed twice from the same prepared state. The record considered task completion, user errors, system errors, assistance, recovery, and preservation of the authoritative source. A critical error would expose an unencrypted original, irreversibly alter the source, encrypt for an unintended recipient without detection, accept an invalid integrity result, or prevent recovery of the intended dataset. A non-critical error would be a reversible action that was blocked or corrected without compromising the source or protected package. The resulting 24 trials are a functional self-evaluation under prepared conditions; they are not an independent participant study and do not provide mobile user-satisfaction, accessibility, or learnability evidence.

Table 3 records how findings from the earlier MapSafe QGIS evaluation were transferred to MapSafe Mobile. *Implemented* denotes a direct mobile control or behaviour; *adapted* denotes a platform-appropriate response rather than literal reproduction; *partial* identifies a remaining convenience or documentation gap; and *not applicable* identifies a desktop-specific concern. These classifications describe design traceability, not participant outcomes. Table 4 combines the success requirements with the outcomes recorded during the first-author effectiveness evaluation. All 24 task repetitions were completed without a user error, system error, critical error, or assistance event.

Table 3.: Traceability from MapSafe QGIS usability findings to MapSafe Mobile design responses.

Heuristic area	**Earlier finding**	**MapSafe Mobile response**	**Status**
Visibility of system status	Retain progress bars and provide prompt feedback after actions.	Safeguard and Access step strips expose the current stage; long-running operations show progress and status text; completion panels name the produced or saved artefact.	Implemented
User control and navigation	Add Back, Forward, Previous, exit, and cancellation controls, particularly around encryption and decryption.	Activities and dialogs provide Back navigation; *Stop* leaves the workflow; *Next* continues; *Remask* and *Bin Again* repeat reversible anonymisation decisions. Busy cryptographic operations disable conflicting controls.	Adapted
Workflow order and error prevention	Explain the order of masking, encryption, and notarisation and why later controls are unavailable until prerequisites are complete.	Compact step indicators show orientation without imposing a compulsory wizard. Safeguard checks for a compatible selected layer at entry, while each feature validates its own prerequisites. Users may begin directly with encryption or verification when appropriate.	Adapted
Recognition rather than recall	Carry masked or binned outputs into encryption and avoid making users repeatedly locate prior inputs.	The artefact selector identifies original, halo-masked, hexagonal-binned, encrypted, and public-key items. Workflow transitions carry the selected output forward; an existing identity is recognised; the owner's key is selected by default; community recipients remain explicit.	Implemented

Continued on next page

Table 3.: Traceability from MapSafe QGIS usability findings to MapSafe Mobile design responses (continued).

Heuristic area	Earlier finding	MapSafe Mobile response	Status
File selection and constraints	Prevent unsupported uploads instead of accepting every file without guidance.	Internal selectors expose recognised MapSafe artefacts and community upload excludes unencrypted originals. Broad Android document-provider selection is retained where MIME filtering previously hid valid `.pgp` files; selected content is subsequently opened and validated with explicit errors.	Adapted
Configuration and secret entry	Validate blockchain fields, apply settings without reloading, and permit optional inspection of obscured secret text.	Network name, chain ID, HTTPS RPC and explorer URLs, and contract-address syntax are validated before use. Saved profiles are read by operational screens without plugin reload. Passphrase and confirmation fields provide eye controls. Wallet private keys are never collected by MapSafe.	Implemented
Error recognition and recovery	Replace generic decryption errors with causes such as an incorrect passphrase.	OpenPGP failures distinguish an incorrect passphrase, a package not addressed to the device's key, an invalid OpenPGP message, missing integrity protection, a failed integrity check, an unknown signer, and an invalid signature.	Implemented
Aesthetic and minimalist design	Preserve a clean interface and avoid controls or explanations that compete with the task and map.	Safeguard and Access share one tabbed launcher; non-functional privacy controls and repeated explanations were removed; the masking range uses one dual-ended control; result cards collapse upward so the map remains visible.	Implemented
Output handling and continuity	Use explicit Save labels, simplify destination selection, preserve progress, and make hashes easy to retain.	Outputs use the predictable `Downloads/MapSafe` folder, show the saved filename, and provide *Open Folder*. Identities and settings persist, while hashes and transaction references are retained with package records. Copy controls are not yet provided for every displayed digest.	Partial
Layer management	Provide a quick way to clear multiple trial anonymisation layers.	Successful decryption clears previously loaded layers and displays only the recovered dataset. A single command for removing every earlier halo or hexagonal candidate has not been added.	Partial
Help and documentation	Link controls to contextual help and provide a concise overview or FAQ.	Headings, concise decision text, stage indicators, confirmation messages, and the planned workflow video provide guidance, but a searchable in-application FAQ and control-level help links remain future work.	Partial
Desktop window behaviour	Fix or minimise the plugin window, confirm closure through Escape, and allow interaction with other QGIS tasks.	MapSafe Mobile uses Android activities and dialogs rather than a resizable desktop plugin window; Android Back, lifecycle restoration, and collapsible map overlays replace these desktop-window behaviours.	Not applicable

Combined effectiveness record

Table 4.: Combined MapSafe Mobile effectiveness task and outcome record from the first-author evaluation. *Rep. 1* and *Rep. 2* report whether the success requirement was met. The error/assistance cell reports user errors and assistance prompts as R1 values followed by R2 values; no system or critical error was recorded.

Task	**Success requirement**	**R1**	**R2**	**Errors/ assist.**	**Outcome**
1. Entry	Stable Safeguard/Access tabs, entry with a compatible point layer, and an immediate understandable warning when spatial input is missing.	Met	Met	0/0; 0/0	Early source confirmation prevents a late incompatible-layer failure.
2. Halo masking	A separate masked layer, an unchanged authoritative source, a visible inverted Spruill score, and sufficient map space to inspect the candidate.	Met	Met	0/0; 0/0	The result and score remain visible while the source is preserved.
3. Hexbinning	A count-bearing polygon layer at the selected H3 resolution without overwriting the original points.	Met	Met	0/0; 0/0	The aggregate is inspected before safeguarding continues.
4. Identity	Matching passphrases supported by visibility controls, no automatic encryption after creation, and recognition of the identity when encryption is reopened.	Met	Met	0/0; 0/0	Eye controls support comparison and identity creation returns to recipient selection.
5. Recipients	Clear distinction between the self-recipient and community recipients, with full-fingerprint acceptance required before the BMA representative can be selected.	Met	Met	0/0; 0/0	The fingerprint requirement prevents an unaccepted community key from being selected.
6. Encryption	One package for the original in `Downloads/MapSafe`, explicit filename confirmation, and recovery by Steven and the BMA representative but not by an unselected identity.	Met	Met	0/0; 0/0	The destination is reported and both selected identities recover the package.
7. Upload	The selected anonymised layer and encrypted package appear in the community, while an unencrypted original is never offered for upload.	Met	Met	0/0; 0/0	Only authorised artefact types are available through community upload.
8. Notarisation	A successful externally approved receipt whose decoded encrypted filename and SHA-256 digest match the local package.	Met	Met	0/0; 0/0	The wallet hand-off and filename–digest confirmation complete without assistance.
9. Download	Selection of the intended `.pgp` package and a downloaded digest that matches its community record.	Met	Met	0/0; 0/0	Community listing reduces filesystem search and downloaded bytes are hash-checked.
10. Verification	A clear filename–digest match or mismatch result and an enabled *Next: Decrypt* action after local hashing.	Met	Met	0/0; 0/0	Verification reports a match and enables decryption.
11. Decryption	Rejection of an incorrect passphrase or non-addressed key, valid integrity and signature reporting, successful recovery, and a disabled decrypt action after success.	Met	Met	0/0; 0/0	The addressed key decrypts successfully and the completed action is disabled.
12. Display	Plaintext saved in `Downloads/MapSafe`, previous layers cleared, and only the newly recovered layer selected and displayed at its extent.	Met	Met	0/0; 0/0	The recovered dataset becomes the sole displayed layer.

Declarations

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Disclosure statement. The authors declare no competing interests.

Ethics approval and consent to participate. This study did not recruit human participants or collect new personal data. The functional evaluation was conducted by the first author using controlled research accounts, synthetic attributes, and previously documented case-study materials. Formal human-participant ethics approval and informed consent were therefore not required under the applicable institutional policy.

Consent for publication. No identifiable participant data are reported. The screenshots use controlled research accounts and role-based identifiers. Where any identifiable account information is retained in a submitted figure, permission for publication will be obtained before submission.

Data availability statement. The source code, APK, performance logs, synthetic test fixtures, derived outputs, and supporting documentation are available through the MapSafe Mobile repository and guide website: `https://osf.io/cyfu7`. Exact North Whangārei coordinates, private keys, passwords, live community credentials, and protected NextGIS resource records are not publicly released because of Indigenous data-sovereignty, privacy, and security considerations. Derived materials may be made available by the corresponding author on reasonable request, subject to approval by the relevant data custodians and community authorities.

Code availability. The MapSafe Mobile source code is openly available at `https://osf.io/cyfu7`. The controlled research APK and its checksum are available through the MapSafe Mobile guide website.

Author contributions. Pankajeshwara Sharma: conceptualisation, methodology, software, investigation, data curation, formal analysis, visualisation, writing—original draft, and writing—review and editing.

Generative AI disclosure. OpenAI Codex based on GPT-5 was used during manuscript preparation for language refinement and organisation of literature. The author retained responsibility for all technical decisions, analyses, measurements, citations, interpretations, and final wording. Generative AI output was not treated as empirical evidence and was not used to generate or alter research data, performance measurements, or evaluation outcomes.

Acknowledgements. The author acknowledges the NextGIS project and the individuals who assisted with platform access, account configuration, and technical testing.

References

Ajayakumar, Jayakrishnan, Andrew J Curtis, and Jacqueline Curtis. 2019. "Addressing the data guardian and geospatial scientist collaborator dilemma: how to share health records for spatial analysis while maintaining patient confidentiality." *International Journal of Health Geographics* 18 (1): 1–12.

Andrachuk, Mark, Melissa Marschke, Charlotte Hings, and Derek Armitage. 2019. "Smartphone Technologies Supporting Community-Based Environmental Monitoring and Implementation: A Systematic Scoping Review." *Biological Conservation* 237: 430–442.

Armstrong, Marc P, Gerard Rushton, and Dale L Zimmerman. 1999. "Geographically masking health data to preserve confidentiality." *Statistics in Medicine* 18 (5): 497–525.

Boukoros, Spyros, Mathias Humbert, Stefan Katzenbeisser, and Carmela Troncoso. 2019. "On (The Lack Of) Location Privacy in Crowdsourcing Applications." In *28th USENIX Security Symposium (USENIX Security 19)*, Santa Clara, CA, 1859–1876. USENIX Association.

Bowen, Judy, and Annika Hinze. 2022. "Participatory data design: managing data sovereignty in IoT solutions." *Interacting with Computers* 34 (2): 60–71.

Briggs, Carolyn, Ingrid Burfurd, Matt Duckham, Olivia Guntarik, Di Kerr, Mark McMillan, and Daisy San Martin Saldias. 2020. "Bridging the geospatial gap: Data about space and indigenous knowledge of place." *Geography Compass* 14 (11): e12542.

Brodsky, Isaac. 2018. "H3: Uber's Hexagonal Hierarchical Spatial Index." Uber Engineering. Open-source hierarchical geospatial indexing system.

Calac, Alec J., and Timothy K. Mackey. 2025. "A Systematic Review of Responsible Stewardship of Research and Health Data from Indigenous Communities." *npj Digital Medicine* 8: 504.

Carroll, Stephanie Russo, Ibrahim Garba, Oscar L. Figueroa-Rodríguez, Jarita Holbrook, Raymond Lovett, Simeon Materechera, Mark Parsons, et al. 2020. "The CARE Principles for Indigenous Data Governance." *Data Science Journal* 19: 43.

Carroll, Stephanie Russo, Edit Herczog, Maui Hudson, Keith Russell, and Shelley Stall. 2021. "Operationalizing the CARE and FAIR Principles for Indigenous Data Futures." *Scientific Data* 8: 108.

Cassa, Christopher A, Shannon C Wieland, and Kenneth D Mandl. 2008. "Re-identification of home addresses from spatial locations anonymized by Gaussian skew." *International journal of health geographics* 7 (1): 1–9.

Chow, Chi-Yin, Mohamed F. Mokbel, and Xuan Liu. 2006. "A Peer-to-Peer Spatial Cloaking Algorithm for Anonymous Location-Based Service." In *Proceedings of the 14th ACM International Symposium on Advances in Geographic Information Systems*, GIS '06, Arlington, Virginia, USA, 171–178. Association for Computing Machinery.

Christin, Delphine. 2016. "Privacy in Mobile Participatory Sensing: Current Trends and Future Challenges." *Journal of Systems and Software* 116: 57–68.

Christin, Delphine, Andreas Reinhardt, Salil S. Kanhere, and Matthias Hollick. 2011. "A Survey on Privacy in Mobile Participatory Sensing Applications." *Journal of Systems and Software* 84 (11): 1928–1946.

Cochrane, Logan, and Jon Corbett. 2020. "Participatory mapping." *Handbook of communication for development and social change* 705–713.

Cormack, Donna, and Tahu Kukutai. 2022. "Indigenous peoples, data, and the coloniality of surveillance." In *New Perspectives in Critical Data Studies: The Ambivalences of Data Power*, 121–141. Springer International Publishing Cham.

Duckham, Matt, and Serene Ho. 2020. "Indigeneity and spatial information science." *Journal of Spatial Information Science* 2020 (21): 71–82.

Engstrom, Teyl, Elton H. Lobo, Kristie Watego, Carmel Nelson, Jinxiang Wang, Howard Wong, Sungkyung Linda Kim, et al. 2024. "Indigenous Data Governance Approaches Applied in Research Using Routinely Collected Health Data: A Scoping Review." *npj Digital Medicine* 7: 68.

Ens, Emilie J. 2012. "Monitoring Outcomes of Environmental Service Provision in Low Socio-

economic Indigenous Australia Using Innovative CyberTracker Technology." *Conservation and Society* 10 (1): 42–52.

Forster, Margaret. 2023. "Amplifying the influence of Māori knowledge in environmental management." *Environment and Planning F* 26349825221133095.

Greenaway, Alison, Sara MacBride-Stewart, Andrea Grant, Susanna Finlay-Smits, Maria Ayala, Will Allen, Liz O'Brien, and Michael Martin. 2023. "Positioning Research to Improve Tree-Biosecurity Relations." *Knowledge Cultures* 11 (1): 234–259.

H3 Project. 2026. "Tables of Cell Statistics Across Resolutions." `https://h3geo.org/docs/core-library/restable/`. Accessed: 6 October 2026.

Hill, Lee, Edward Ashby, Nick Waipara, Robin Taua-Gordon, Aleesha Gordon, Fredrik Hjelm, Stanley E Bellgard, Emma Bodley, and Linley K Jesson. 2021. "Cross-cultural leadership enables collaborative approaches to management of Kauri dieback in Aotearoa New Zealand." *Forests* 12 (12): 1671.

Hojati, Majid, Carson Farmer, Rob Feick, and Colin Robertson. 2021. "Decentralized geoprivacy: leveraging social trust on the distributed web." *International Journal of Geographical Information Science* 35 (12): 2540–2566.

Hojati, Majid, Rob Feick, Steven Roberts, Carson Farmer, and Colin Robertson. 2023. "Distributed Spatial Data Sharing: A New Model for Data Ownership and Access Control." *Journal of Spatial Information Science* (27): 1–26.

ISO. 2018. "Ergonomics of human-system interaction–part 11: Usability: Definitions and concepts." *ISO 9241* 11.

Jennings, Lydia, Talia Anderson, Andrew Martinez, Rogena Sterling, Dominique David Chavez, Ibrahim Garba, Maui Hudson, Nanibaa' A. Garrison, and Stephanie Russo Carroll. 2023. "Applying the CARE Principles for Indigenous Data Governance to Ecology and Biodiversity Research." *Nature Ecology & Evolution* 7: 1547–1551.

Keßler, Carsten, and Grant McKenzie. 2018. "A geoprivacy manifesto." *Transactions in GIS* 22 (1): 3–19.

Kidd, Dorothy. 2019. "Extra-activism: Counter-mapping and data justice." *Information, Communication & Society* 22 (7): 954–970.

Kim, Jong Wook, Kennedy Edemacu, and Beakcheol Jang. 2022. "Privacy-Preserving Mechanisms for Location Privacy in Mobile Crowdsensing: A Survey." *Journal of Network and Computer Applications* 200: 103315.

Kukutai, Tahu, and Donna Cormack. 2019. "Mana Motuhake ā-Raraunga: Datafication and Social Science Research in Aotearoa." *Kōtuitui: New Zealand Journal of Social Sciences Online* 14 (2): 201–208.

Kukutai, Tahu, and John Taylor, eds. 2016. *Indigenous Data Sovereignty: Toward an Agenda*. Vol. 38 of *CAEPR Research Monograph*. Canberra, Australia: ANU Press.

Kwusen Research & Media. 2026. "GeoKeeper." `https://kwusen.ca/geokeeper`. Accessed: 7 August 2026.

Lambert, Simon, Nick Waipara, Amanda Black, Melanie Mark-Shadbolt, and Waitangi Wood. 2018. "Indigenous biosecurity: Māori responses to kauri dieback and myrtle rust in Aotearoa New Zealand." *The human dimensions of forest and tree health: Global perspectives* 109–137.

Lovett, Raymond, Vanessa Lee, Tahu Kukutai, Donna Cormack, Stephanie Rainie, and Jennifer Walker. 2019. "Good Data Practices for Indigenous Data Sovereignty and Governance." In *Good Data*, edited by Angela Daly, S. Kate Devitt, and Monique Mann, 26–36. Amsterdam: Institute of Network Cultures.

Lyver, P. O'. B., P. Timoti, C. J. Jones, S. J. Richardson, B. L. Tahi, and S. Greenhalgh. 2017. "An Indigenous Community-Based Monitoring System for Assessing Forest Health in New Zealand." *Biodiversity and Conservation* 26: 3183–3212.

Marriott, Rhonda, Rita Farrell, Juli Coffin, Tracy Reibel, and Roz Walker. 2025. "Recognising Indigenous Data Sovereignty and Implementing Indigenous Data Governance at the Ngangk Yira Institute for Change." *First Nations Health and Wellbeing – The Lowitja Journal* 3: 100072.

Masinde, Brian K., Caroline M. Gevaert, Michael H. Nagenborg, and Jaap A. Zevenbergen.

2023. “Group-Privacy Threats for Geodata in the Humanitarian Context.” *ISPRS International Journal of Geo-Information* 12 (10): 393.

Massarella, Kate, Susannah M Sallu, and Jonathan E Ensor. 2020. “Reproducing injustice: Why recognition matters in conservation project evaluation.” *Global Environmental Change* 65: 102181.

McKenzie, Grant, Carsten Keßler, and Clio Andris. 2019. “Geospatial privacy and security.” *Journal of spatial information science* (19): 53–55.

Meftah, Lakhdar, Romain Rouvoy, and Isabelle Chrisment. 2021. “Empowering Mobile Crowdsourcing Apps with User Privacy Control.” *Journal of Parallel and Distributed Computing* 147: 1–15.

NextGIS. 2026a. “Managing Users.” `https://docs.nextgis.com/docs_ngweb/source/users.html`. NextGIS Web documentation; accessed: 6 October 2026.

NextGIS. 2026b. “Permission System in NextGIS Web.” `https://docs.nextgis.com/docs_ngweb/source/permissions.html`. NextGIS Web documentation; accessed: 6 October 2026.

Nielsen, Jakob. 1995. “How to conduct a heuristic evaluation.” *Nielsen Norman Group* 1 (1): 8.

Olson, Rachel, Jeffrey Hackett, and Steven DeRoy. 2016. “Mapping the digital terrain: towards indigenous geographic information and spatial data quality indicators for indigenous knowledge and traditional land-use data collection.” *The Cartographic Journal* 53 (4): 348–355.

Paul, Sophie A. L., A. Meriwether W. Wilson, Rachide Cachimo, and Michael A. Riddell. 2016. “Piloting Participatory Smartphone Mapping of Intertidal Fishing Grounds and Resources in Northern Mozambique: Opportunities and Future Directions.” *Ocean & Coastal Management* 134: 79–92.

Pyrgelis, Apostolos, Emiliano De Cristofaro, and Gordon Ross. 2016. “Privacy-Friendly Mobility Analytics Using Aggregate Location Data.” In *Proceedings of the 24th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, SIGSPATIAL ’16, 34:1–34:10. Association for Computing Machinery.

Rainie, Stephanie Carroll, Tahu Kukutai, Maggie Walter, Oscar Luis Figueroa-Rodríguez, Jennifer Walker, and Per Axelsson. 2019. “Indigenous Data Sovereignty.” In *The State of Open Data: Histories and Horizons*, edited by Tim Davies, Stephen B. Walker, Mor Rubinstein, and Fernando Perini, 300–319. Cape Town and Ottawa: African Minds and International Development Research Centre.

Rao, Jinmeng, Song Gao, Mingxiao Li, and Qunying Huang. 2021. “A privacy-preserving framework for location recommendation using decentralized collaborative machine learning.” *Transactions in GIS* 25 (3): 1153–1175.

Reid, Geneviève, and Renée E Sieber. 2022. “Learning from critiques of GIS for assessing the geoweb and indigenous knowledges.” *GeoJournal* 87 (2): 875–893.

Reown. 2026. “WalletKit Overview for Android.” `https://docs.reown.com/walletkit/overview?platform=android`. Accessed: 6 October 2026.

Reyes-García, Victoria, Álvaro Fernández-Llamazares, Yildiz Aumeeruddy-Thomas, Petra Benyei, Rainer W Bussmann, Sara K Diamond, David García-Del-Amo, et al. 2022. “Recognizing Indigenous peoples’ and local communities’ rights and agency in the post-2020 Biodiversity Agenda.” *Ambio* 51 (1): 84–92.

Reyes-García, Victoria, Adrien Tofighi-Niaki, Beau J. Austin, Petra Benyei, Finn Danielsen, Álvaro Fernández-Llamazares, Aditi Sharma, Ramin Soleymani-Fard, and Maria Tengö. 2022. “Data Sovereignty in Community-Based Environmental Monitoring: Toward Equitable Environmental Data Governance.” *BioScience* 72 (8): 714–717.

Sahr, Kevin, Denis White, and A. Jon Kimerling. 2003. “Geodesic Discrete Global Grid Systems.” *Cartography and Geographic Information Science* 30 (2): 121–134.

Sharma, Pankajeshwara, Michael Martin, and David Swanlund. 2023. “MapSafe: A Complete Tool for Achieving Geospatial Data Sovereignty.” *Transactions in GIS* 27 (6): 1680–1698.

Sharma, Pankajeshwara, Michael Martin, David Swanlund, Cecilia Latham, Dean Anderson, and Waitangi Wood. 2024. “A Cloud-Based Solution for Trustless Indigenous Data

Sovereignty: Protecting Māori Biodiversity Management Data in Aotearoa New Zealand." *Transactions in GIS* 28 (4): 836–857.

Sharma, Pankajeshwara, Nick Young, and Taras Dubrava. 2025. "MapSafe QGIS plugin: a complete open-source geoprivacy tool for desktop GIS applications." *Journal of Location Based Services* 19 (4): 287–322.

Sporle, Andrew, Maui Hudson, and Kiri West. 2020. "Indigenous data and policy in Aotearoa New Zealand." *Indigenous data sovereignty and policy* 62–80.

Swanlund, David, Nadine Schuurman, and Mariana Brussoni. 2020. "MaskMy. XYZ: An easy-to-use tool for protecting geoprivacy using geographic masks." *Transactions in GIS* 24 (2): 390–401.

Te Mana Raraunga. 2018. *Principles of Māori Data Sovereignty*. Technical Report. Te Mana Raraunga Māori Data Sovereignty Network.

Wainwright, Joel, and Joe Bryan. 2009. "Cartography, territory, property: postcolonial reflections on indigenous counter-mapping in Nicaragua and Belize." *Cultural geographies* 16 (2): 153–178.

Waipara, NW, S Hill, LMW Hill, EG Hough, and IJ Horner. 2013. "Surveillance methods to determine tree health distribution of kauri dieback disease and associated pathogens." *New Zealand Plant Protection* 66: 235–241.

Walter, Maggie, Raymond Lovett, Bobby Maher, Bhiamie Williamson, Jacob Prehn, Gawaian Bodkin-Andrews, and Vanessa Lee. 2021. "Indigenous data sovereignty in the era of big data and open data." *Australian Journal of Social Issues* 56 (2): 143–156.

Wilson, Nicole J. 2025. "Does Community-Based Monitoring Advance Indigenous Self-Determination? Inuit-Led Monitoring and Governance in Nunavut and Greenland." *Environmental Science & Policy* 172: 104215.

Wiseman, Nathanael D., and Douglas K. Bardsley. 2016. "Monitoring to Learn, Learning to Monitor: A Critical Analysis of Opportunities for Indigenous Community-Based Monitoring of Environmental Change in Australian Rangelands." *Geographical Research* 54 (1): 52–71.

Wouters, Paul, Daniel Huigens, Justus Winter, and Yutaka Niibe. 2024. *OpenPGP*. RFC 9580. Internet Engineering Task Force.

Ye, Hang, Kai Han, Chaoting Xu, Jingxin Xu, and Fei Gui. 2019. "Toward Location Privacy Protection in Spatial Crowdsourcing." *International Journal of Distributed Sensor Networks* 15 (3): 1550147719830568.

Zhang, Hongyu, and Grant McKenzie. 2022. "Rehumanize geoprivacy: from disclosure control to human perception." *GeoJournal* 1–20.

Zhao, Pengxiang, Jesus Rodrigo Cedeno Jimenez, Maria Antonia Brovelli, and Ali Mansourian. 2022. "Towards Geospatial Blockchain: A Review of Research on Blockchain Technology Applied to Geospatial Data." *AGILE GIScience Series* 3: 71.

Zimmerman, Dale L, and Claire Pavlik. 2008. "Quantifying the effects of mask metadata disclosure and multiple releases on the confidentiality of geographically masked health data." *Geographical analysis* 40 (1): 52–76.